

Econ 139: Labor Economics

Monopsony (Empirics)

David Arnold





October 15 — Write down your first and last name

Empirical Evidence for Monopsony Power

- In the last section we discussed why thinking about labor markets as monopsonistic might be important, but we didn't provide any empirical evidence
- This section will focus on a few different types of evidence
 - Large literature estimating elasticity of labor supply to the firm
 - Range of observations that appear inconsistent with perfect competition, but make sense in monopsonistic models
 - Case studies exploring collusion in labor markets

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Elasticity of Labor Supply to the Firm

- The Elasticity of Labor Supply to the firm measures how responsive individuals are to changes in wages
- In practice, it is extremely difficult to measure
- The reason why is not just specific to the elasticity of labor supply, but plagues all aspects of economics research
 - Again it really comes down to a case of confounding variables, something that plagues all of science

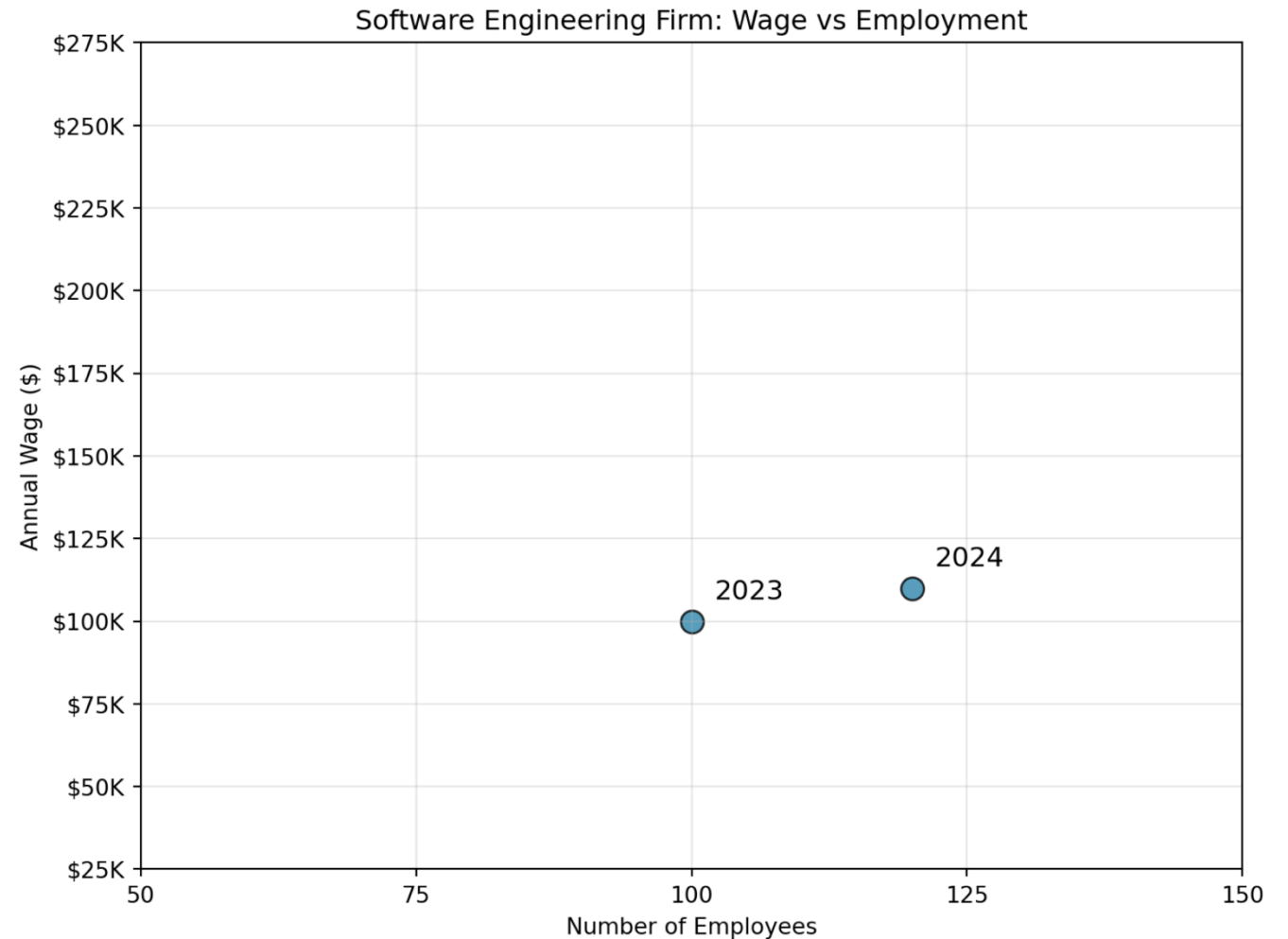


A Concrete Example

- Let's imagine a software firm had 100 employees in 2023 and paid an average salary of \$100,000
- In 2023, they pay an average salary of \$110,000 and grow to 120 employees
- Let's see what this data would look like

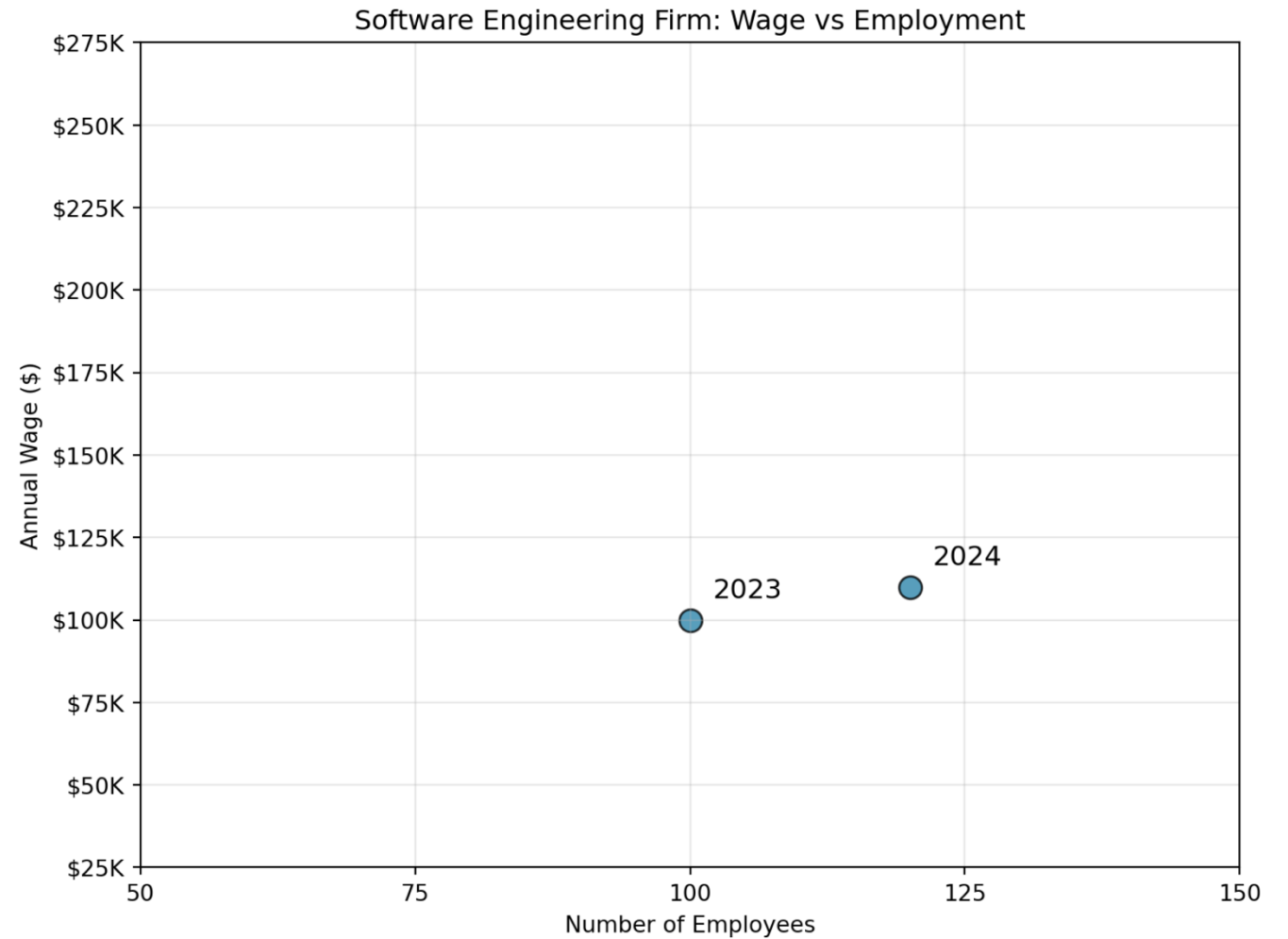
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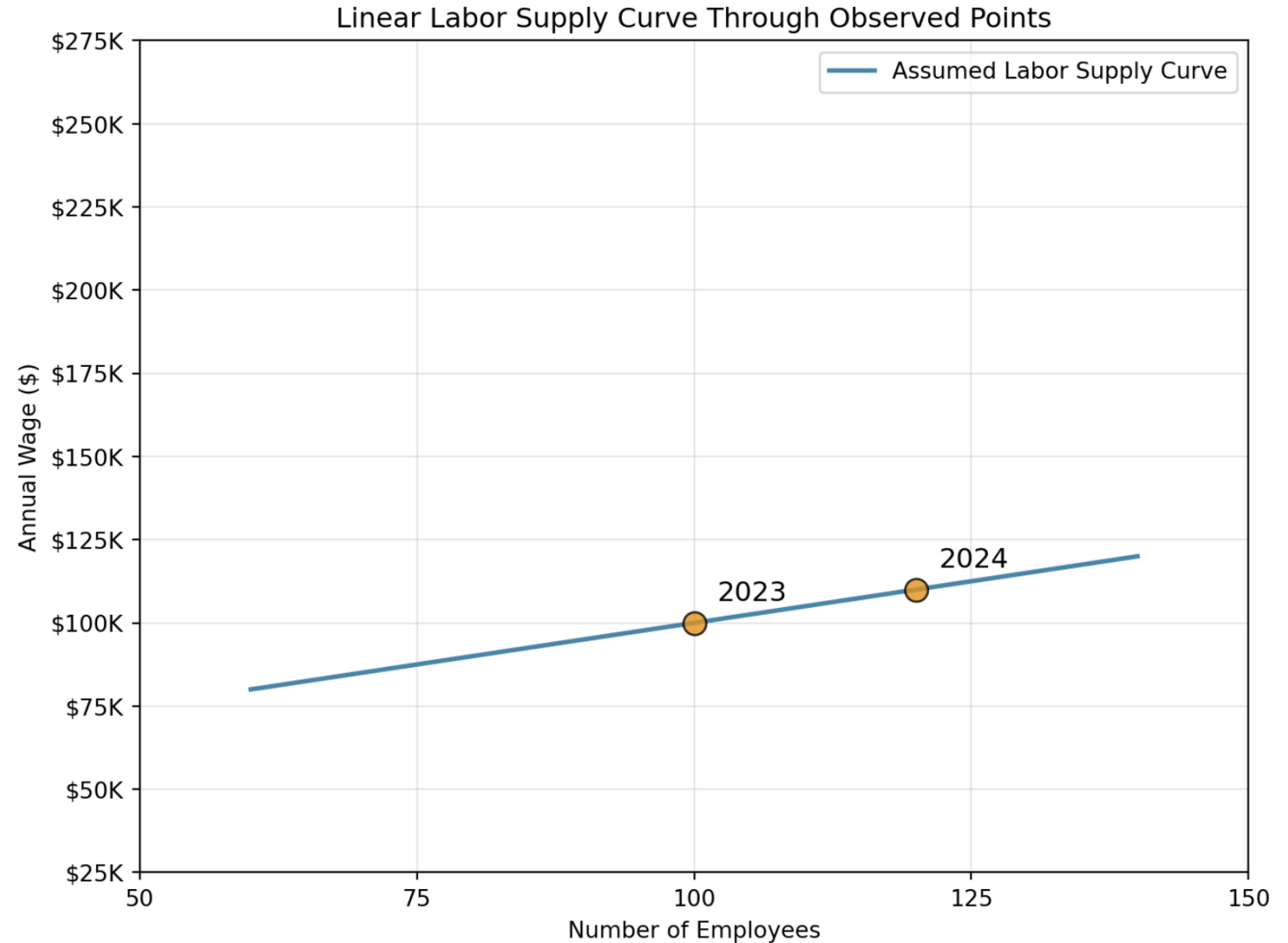
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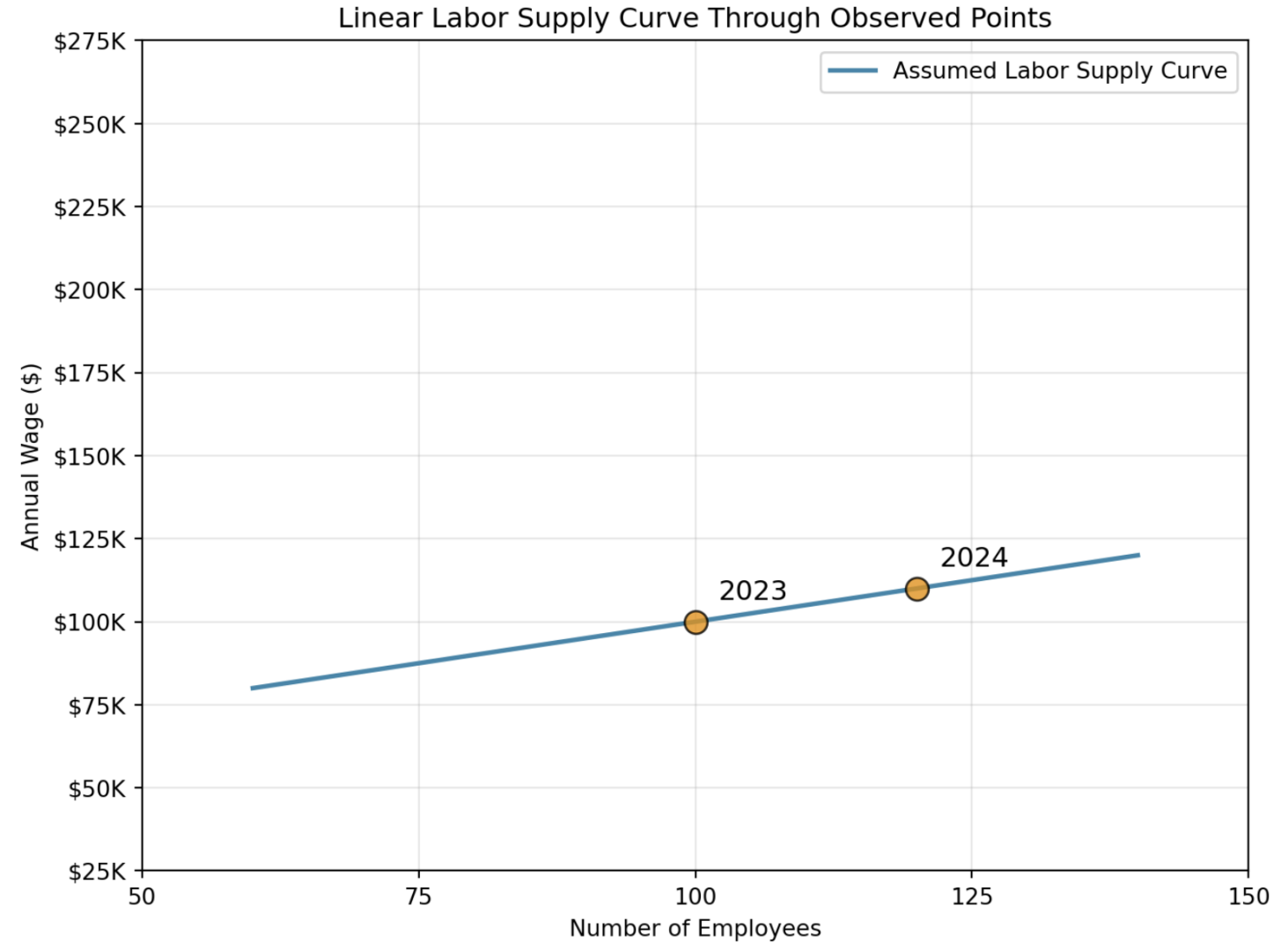
- It might be tempting to connect these two points and assume this is the labor supply curve
- We can also compute the elasticity of labor supply

$$\eta = \frac{\% \Delta L}{\% \Delta W} = \frac{20\%}{10\%} = 2$$



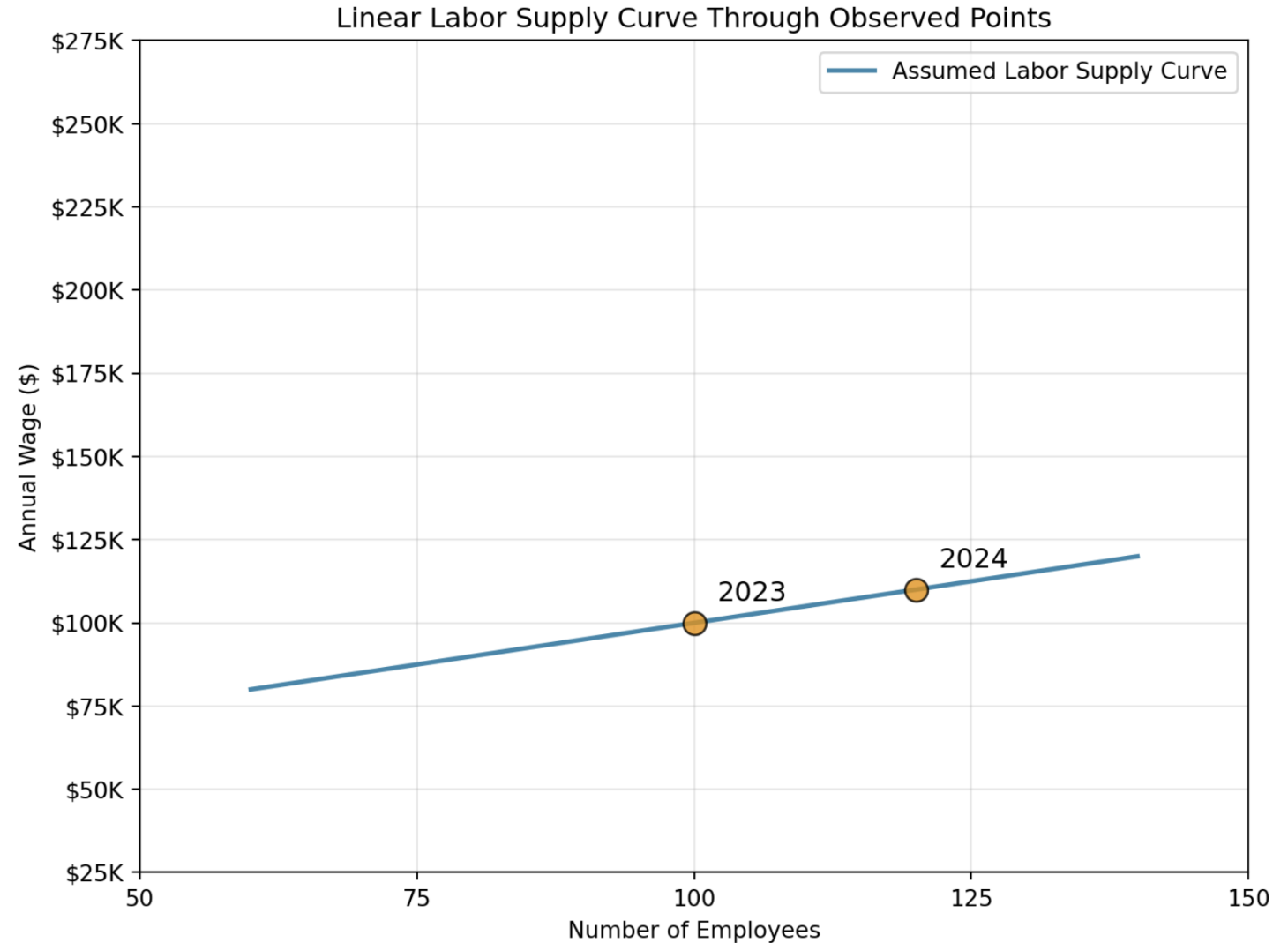
A Concrete Example

- But is this necessarily right?
- Let's think about what determines equilibrium employment

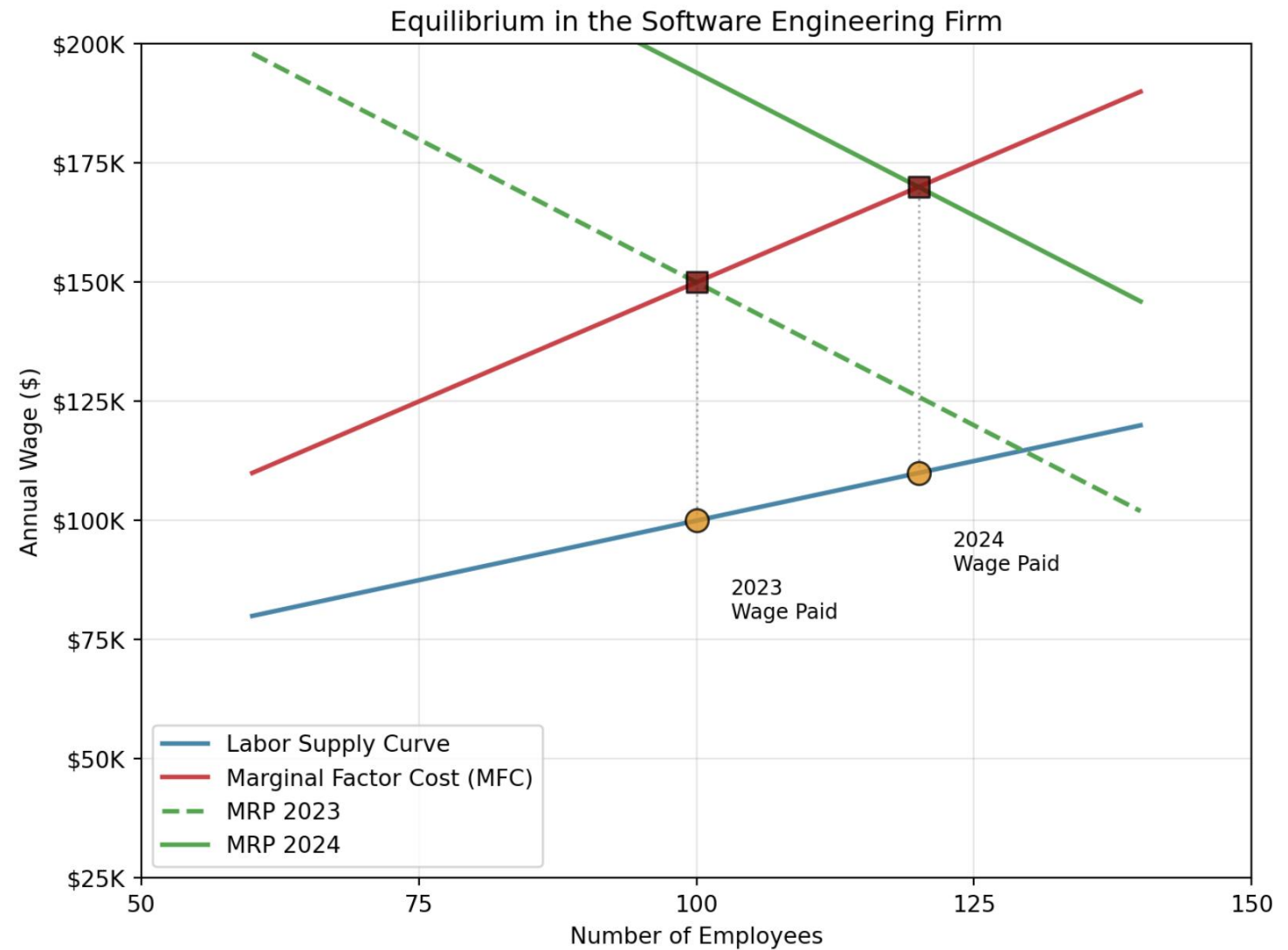


A Concrete Example

- But is this necessarily right?
- Let's think about what determines equilibrium employment
 - It's the point where MRPL intersects the marginal factor cost curve
- So let's add some MRPL and MFC cost curves that are consistent with this data

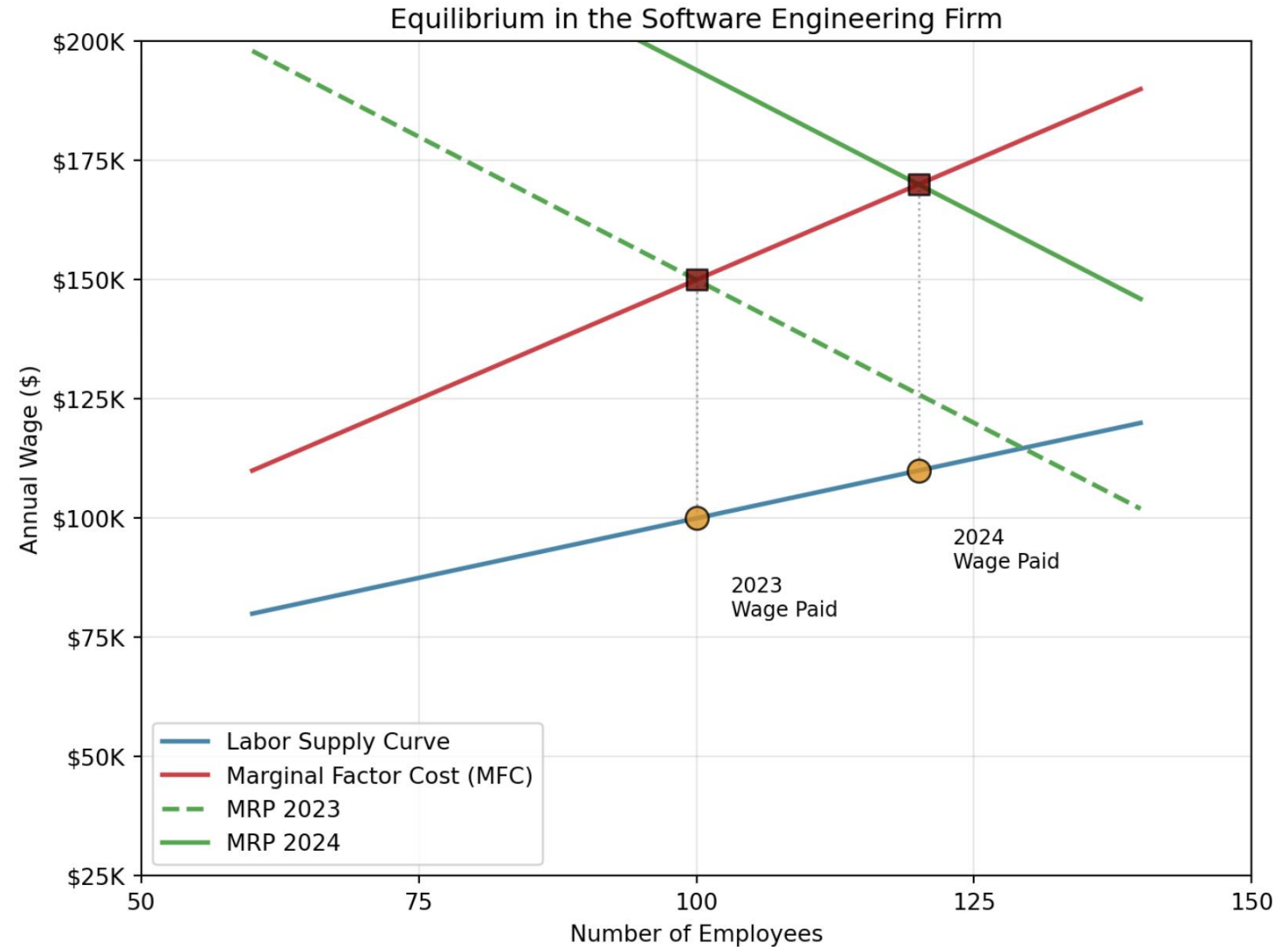


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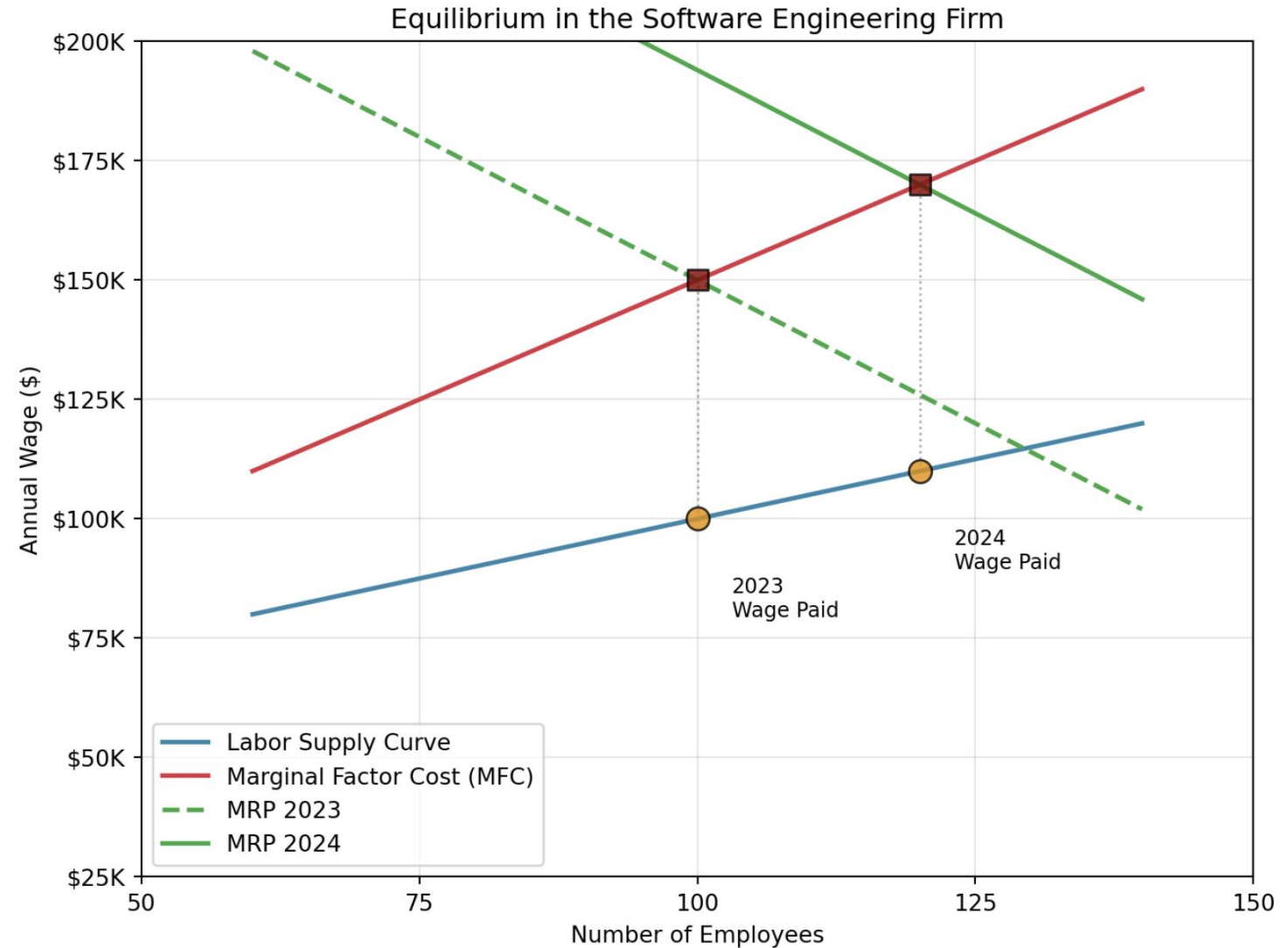
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- One thing you may notice: the MRP curves (Marginal revenue product of labor) shifted out
 - This is the only way we can sustain these points as equilibrium assuming the MRP curves are downward sloping



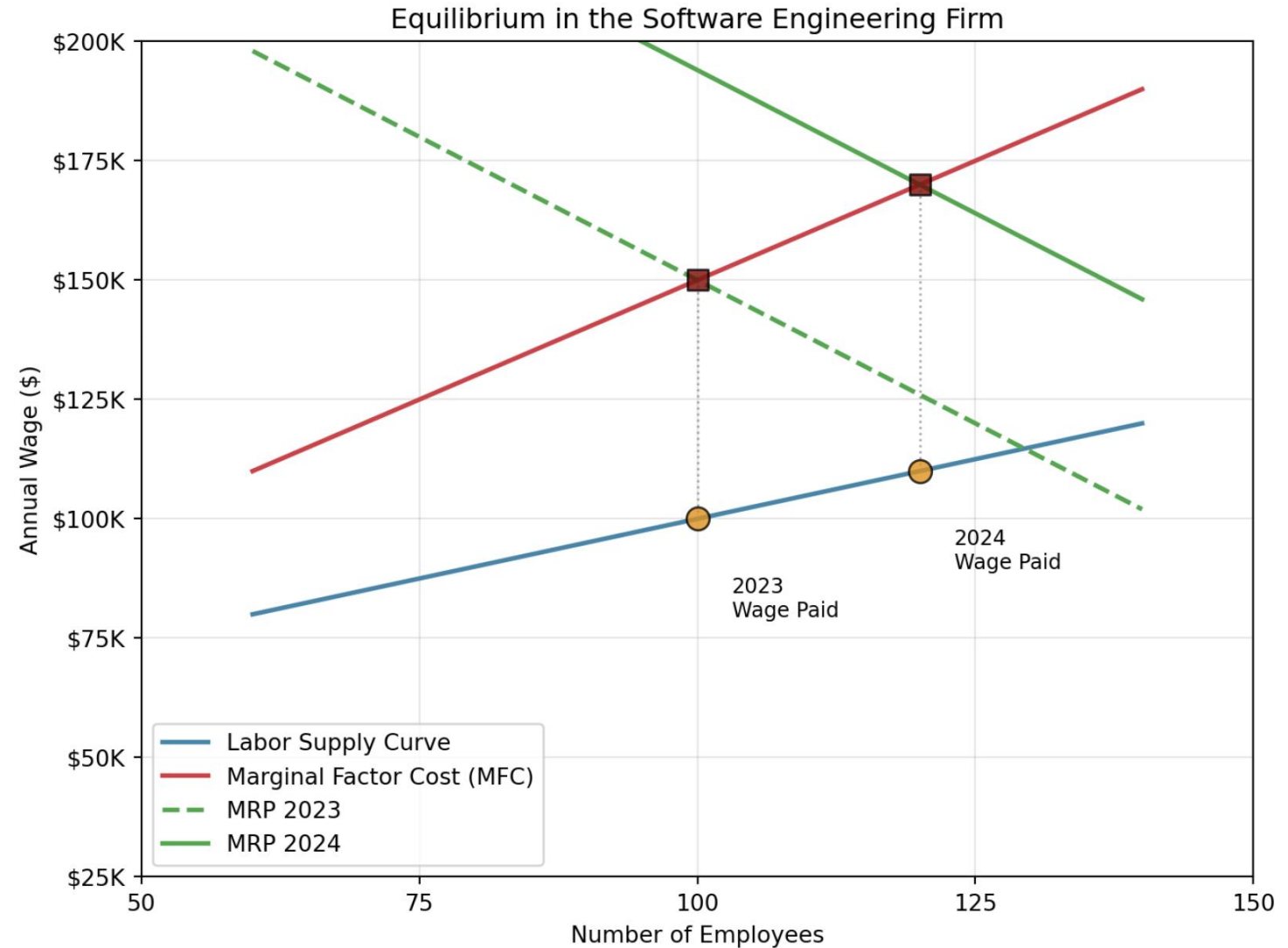
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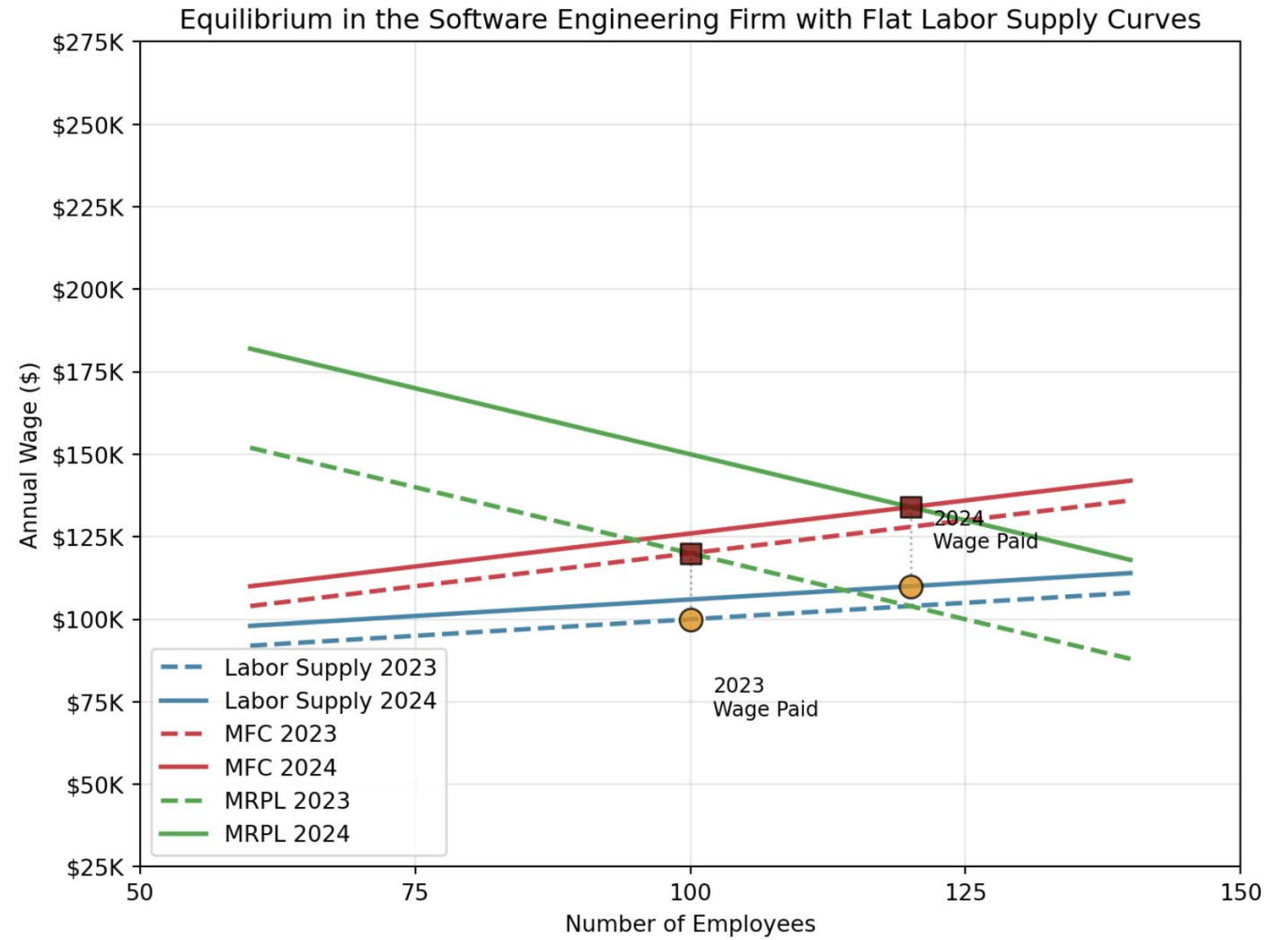


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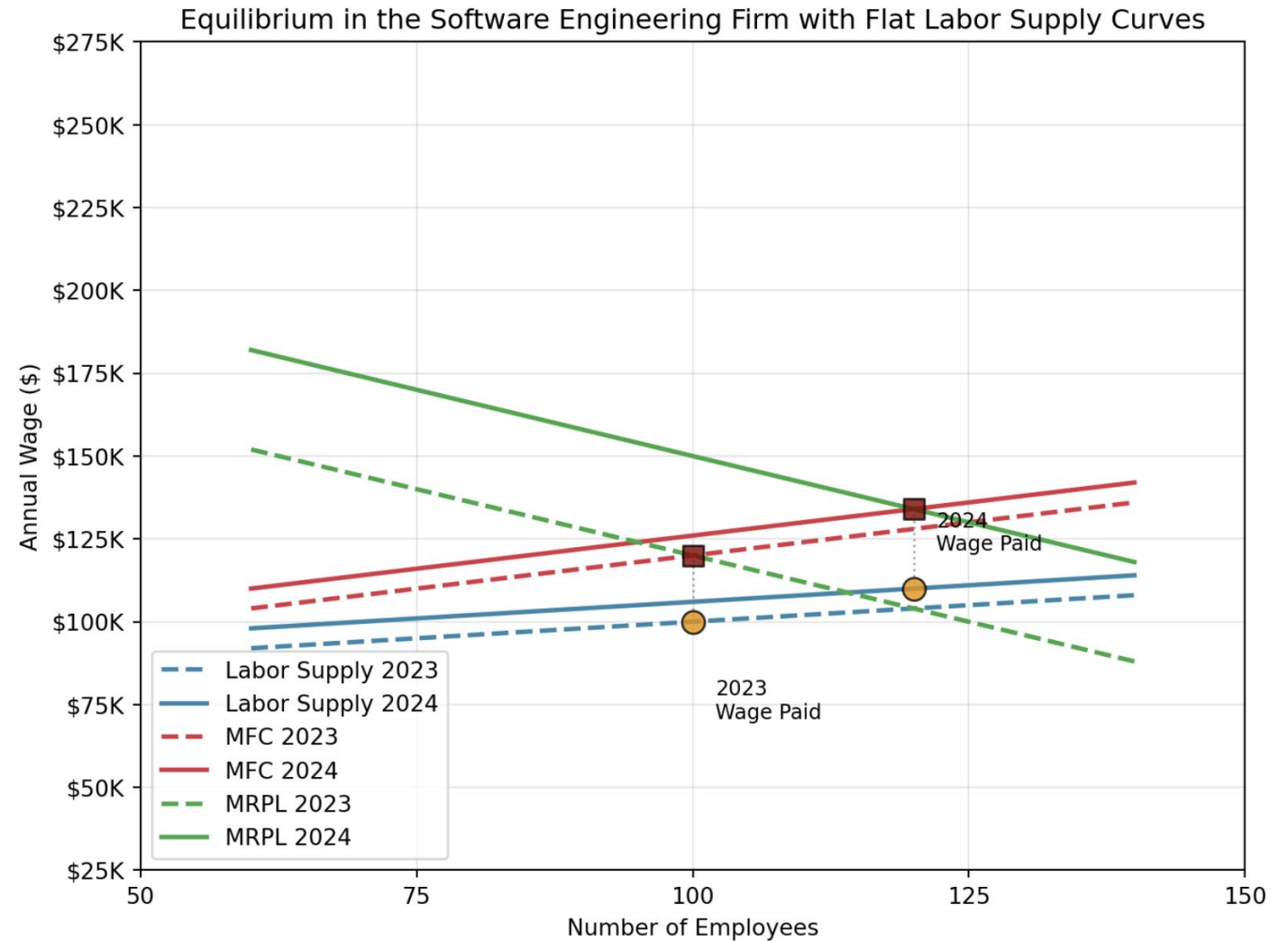


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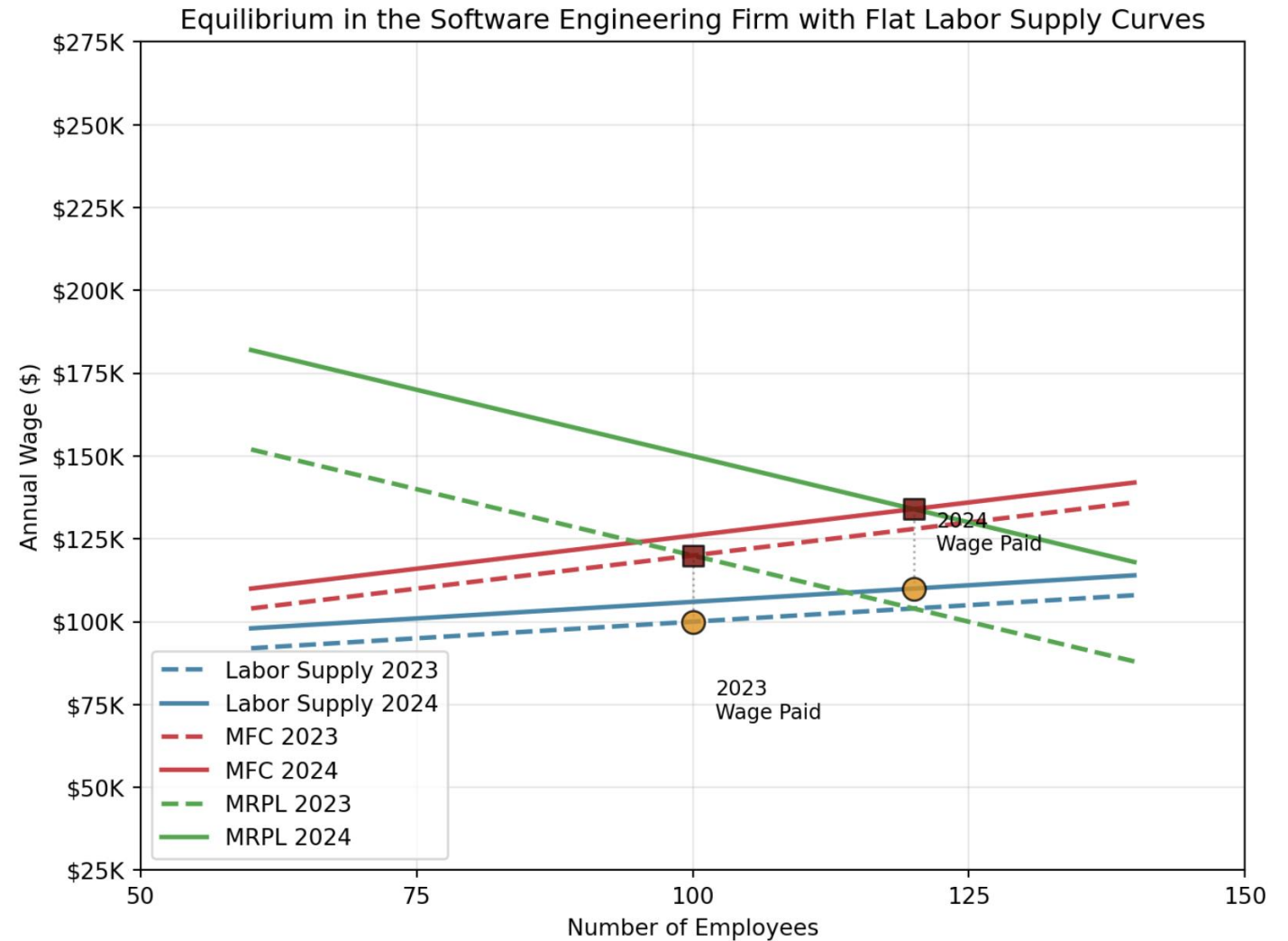
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- These curves are quite flat
- This implies small changes in w result in large changes in L
 - So the elasticity of labor supply is very large and the firm has very little monopsony power



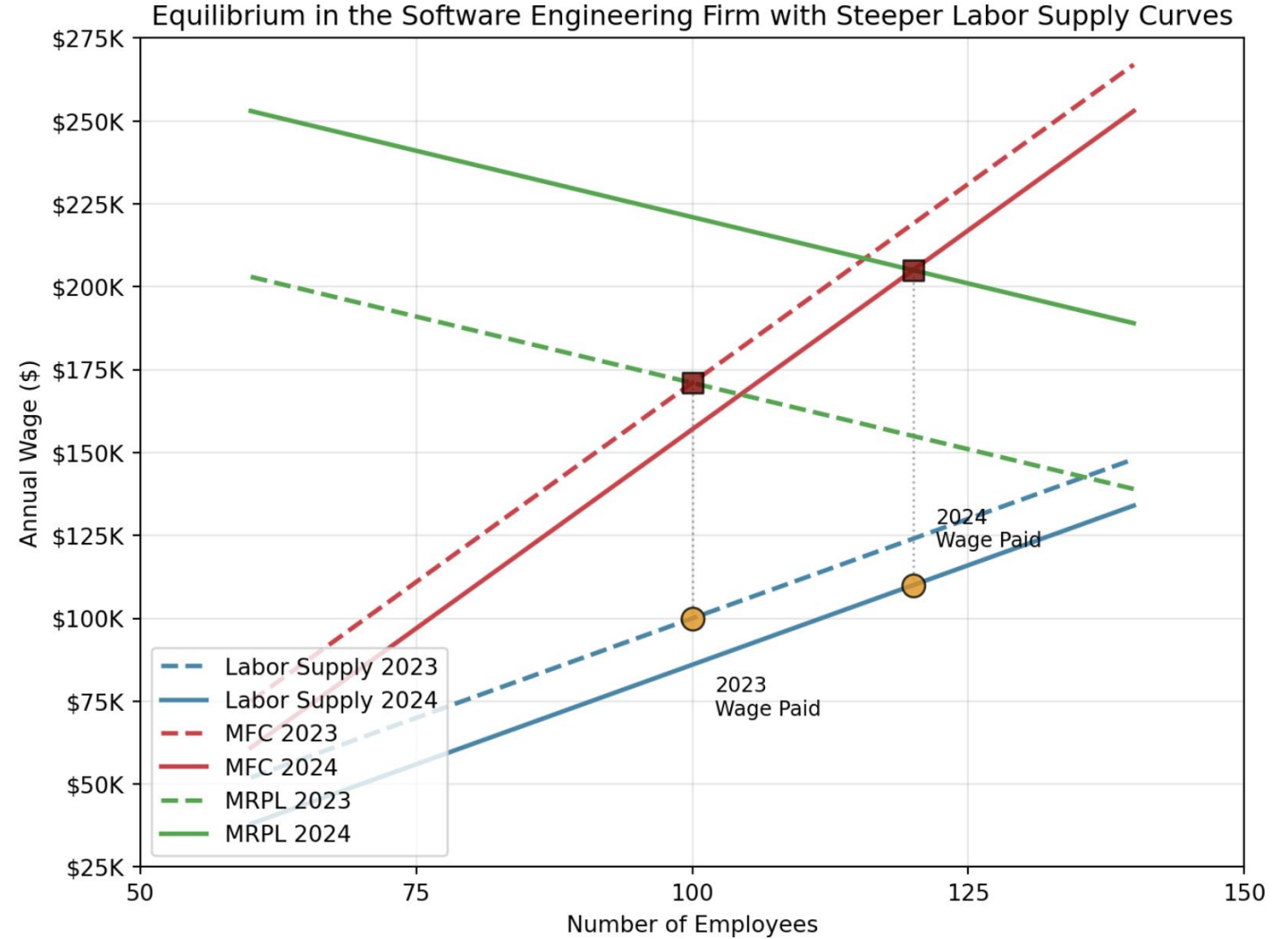
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- These curves are quite flat
- This implies small changes in w result in large changes in L
 - So the elasticity of labor supply is very large and the firm has very little monopsony power
- But now that we can shift **both MRP and the labor supply** curve we have a problem



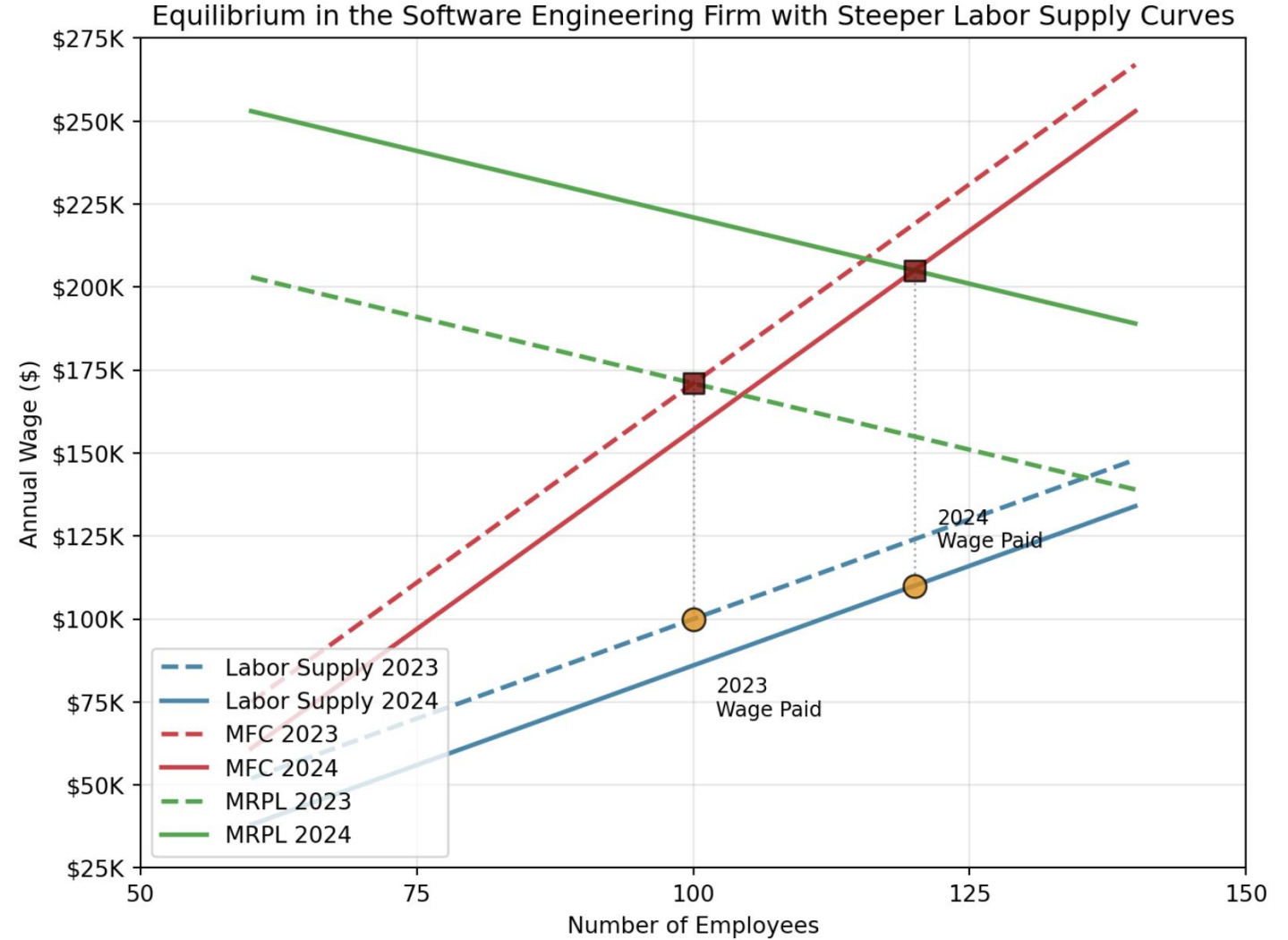
A Concrete Example

- These curves are also consistent with the data

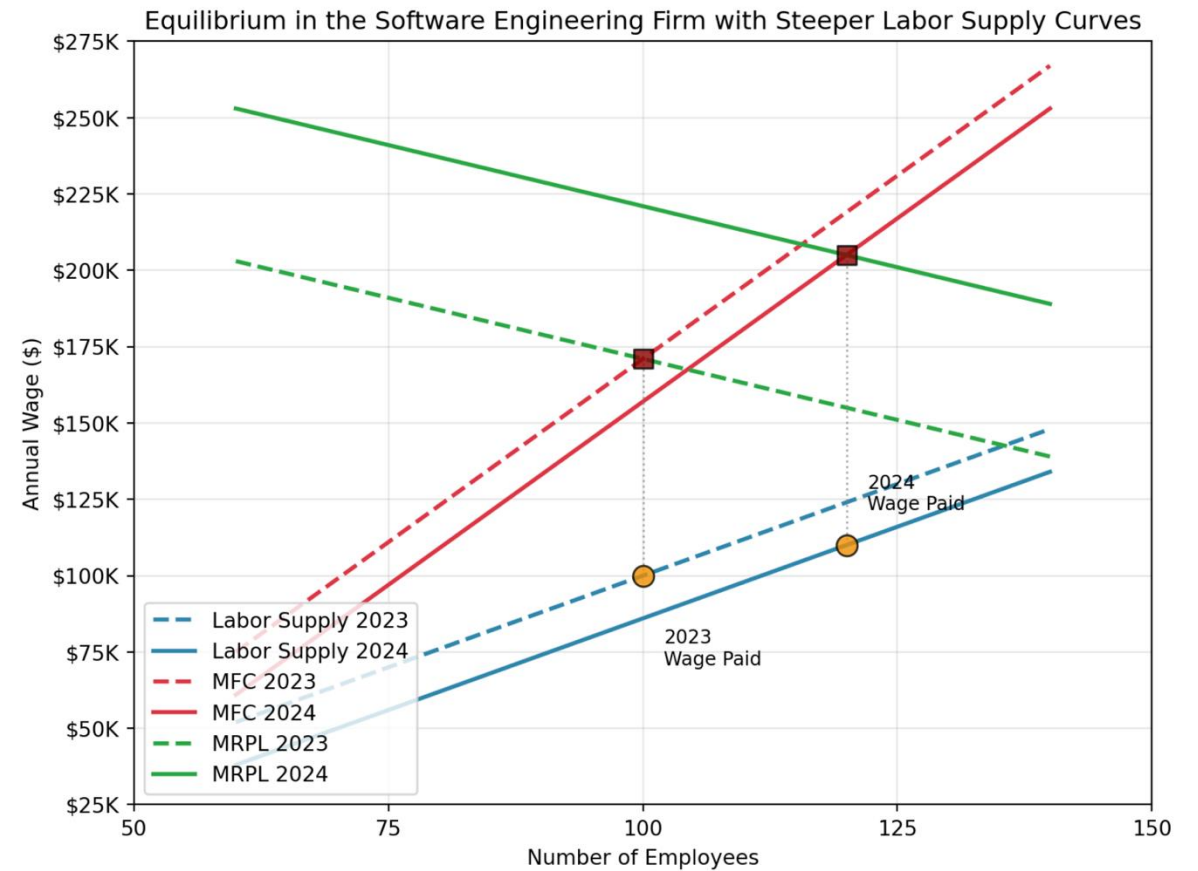
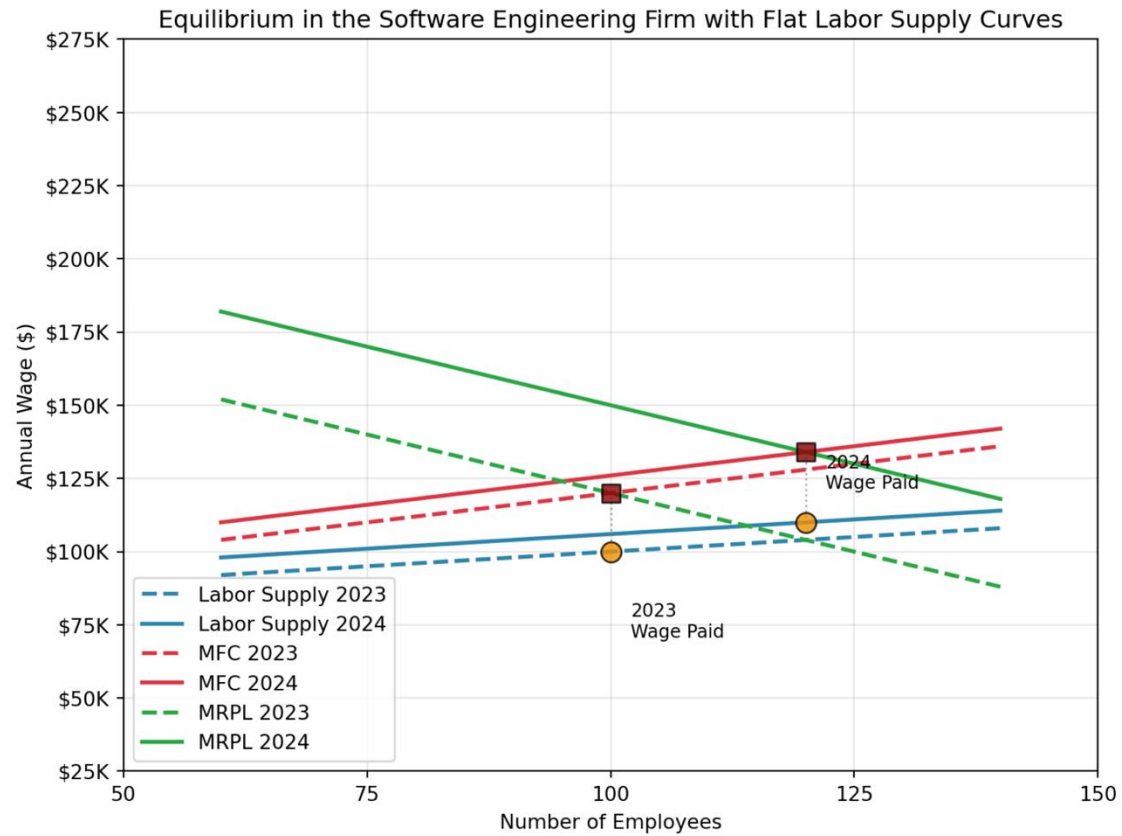


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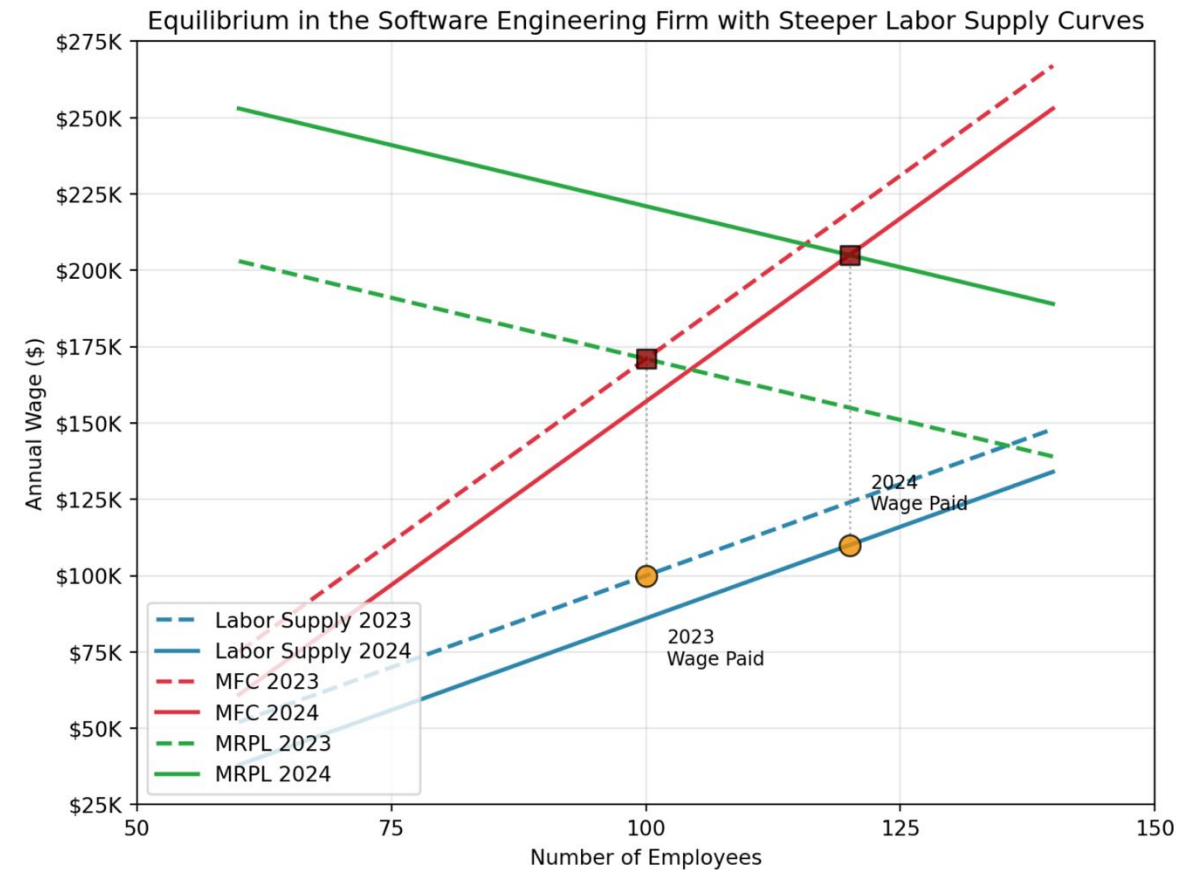
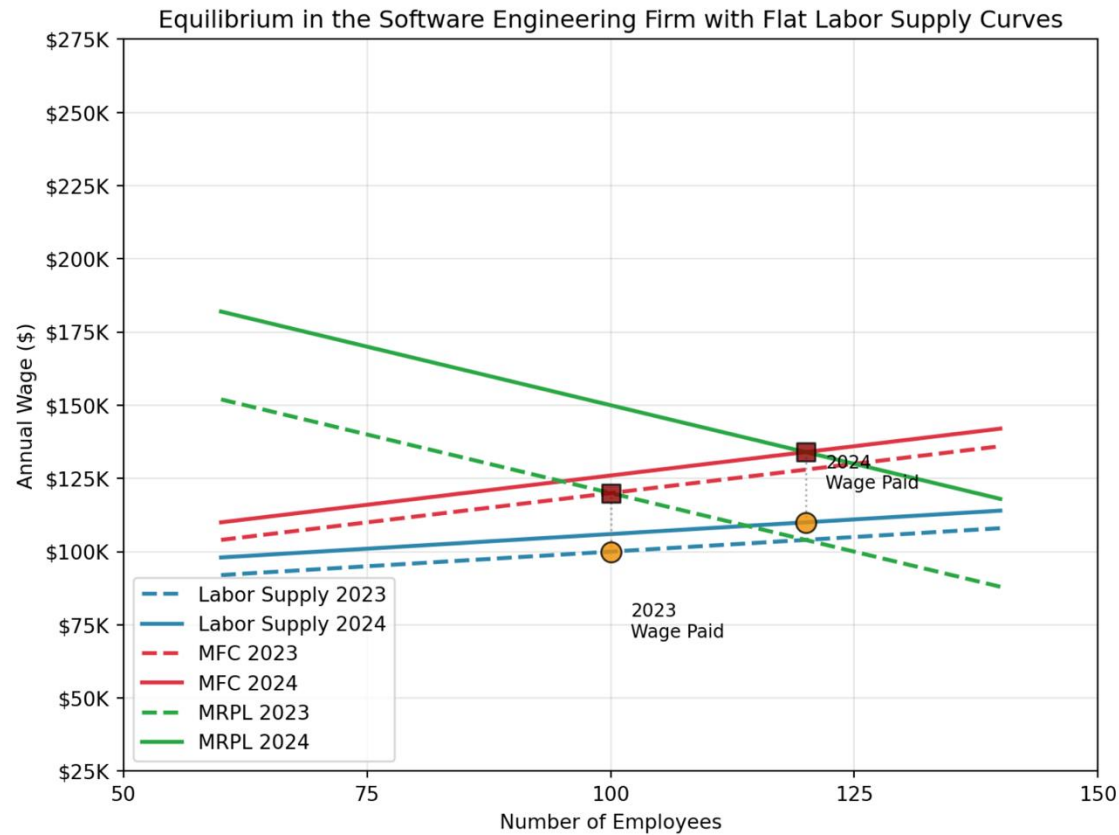
- These curves are also consistent with the data
- But they are steeper, implying more monopsony power for the firm relative to the previous graph



The Problem



The Problem



Both consistent with the data, but imply very different elasticities of labor supply

The (General) Problem

- When both labor supply and MRPL can change, no way to use wages and employment to estimate the elasticity of labor supply
- This is a deep general problem (also an issue in estimating product demand and supply curves)
- The general solution is to find an instrument (you will learn about these in further econometrics courses). Basically, ensures only one curve is moving at a time
- In our setting, if we could find an **exogenous** wage increase (i.e. one not driven by changing MRPL or supply), then we might be able to estimate the elasticity of labor supply at the firm level

Exogenous wage changes

- The word **exogenous** has a precise definition in econometrics. More colloquially, it means something that comes from outside the model
- The best source of exogenous variation is something that is random
 - A famous example is studying the impact of being a veteran on long-term earnings. The Vietnam Era Draft in the U.S. generated exogenous variation in who served in the military
- So in our setting, we would like something that increases (or decreases) wages, but for a “random” reason not related to the firm’s decision

Paper	Type of Study	Exogenous Variation	Estimate (Firm-level elasticity of labor supply)
Staiger, Spetz, and Phibbs (2010)	Quasi-experimental	Mandates increase in wages at VA hospitals	0.1
Falch (2010)	Quasi-experimental	In Norway, a subset of schools received a wage increase	3.3
Del Bo, Finan and Rossi (2016)	Experimental	Government in Mexico randomized wages across different recruitment sites	2.15
Dube, Jacobs, Naidu and Suri	Experimental	M Turk is an online labor platform. Randomized wages across jobs	0.1

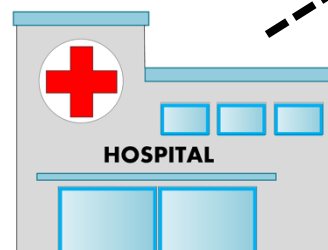


VA Hospital increases wage





Do nearby hospitals react?



VA Hospital increases wage

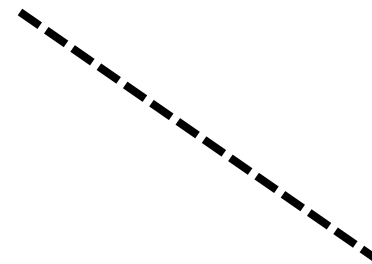




How about farther away hospitals?



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TABLE V
THE EFFECTS OF FINANCIAL INCENTIVES ON RECRUITMENT

	Accepted (1)	Accepted (2)	Rejected (3)	Not reachable (4)
High-wage offer	0.151 [0.054]***	0.160 [0.054]**	−0.017 [0.034]	−0.135 [0.054]***
Characteristics				
Male		−0.080 [0.058]		
Years of schooling		−0.022 [0.008]**		
High IQ		−0.017 [0.053]		
Wage in previous job > 5,000 pesos		−0.007 [0.067]		
Big 5 index		0.042 [0.038]		
PSM index		−0.024 [0.044]		
Mean of dependent variable	0.55	0.55	0.13	0.32
Observations	350	343	350	350
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Using a theoretical model, they convert this estimate to a labor supply elasticity of 2.15

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Azar and Marinescu (2024)

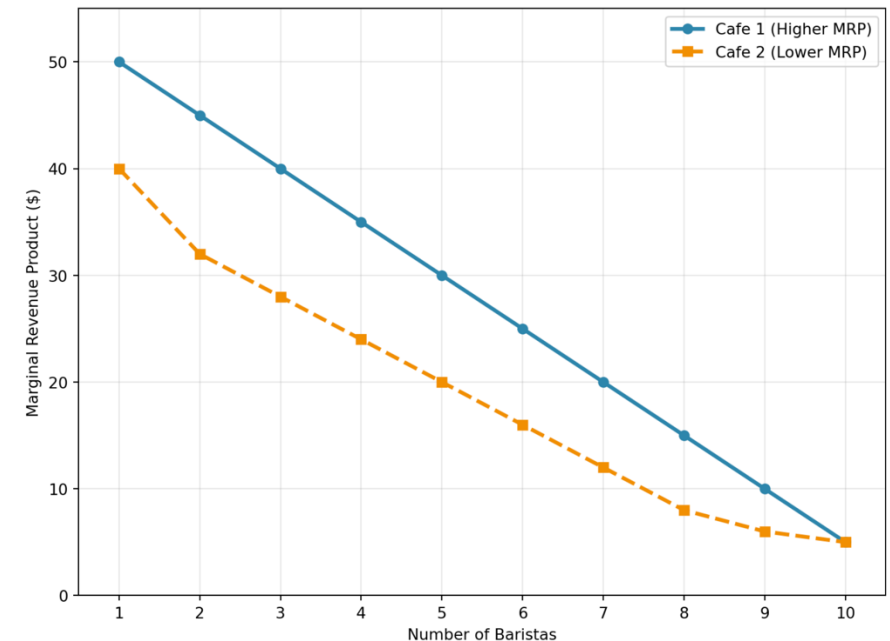
- Azar and Marinescu (2024) provide a review of recent work studying monopsony power
- Most estimates are in the range of 1-4
- Using some standard models, this implies wages are between 15%-50% lower than marginal productivity of the worker

Other Evidence

- We are now going to dive into three different sets of evidence that make “more sense” in imperfectly competitive models of the labor market
 - Firm-specific pay premia
 - Institutional boundaries to mobility
 - Concentration in labor markets

Firm-Specific Pay Premia

- In an earlier class we studied two cafes in a perfectly competitive model
- While they had different productivity, they paid the exact same wage in equilibrium
 - Café 2 hired fewer workers, but the wage paid is exactly the same, as it has to be in a perfectly competitive model
- But in reality, don't firms pay different wages?





October 17 — Write down your first and last name



Imagine we have the two cafes below. In a perfectly competitive labor market, cafe 1 will pay...

Firm-Specific Pay Premia

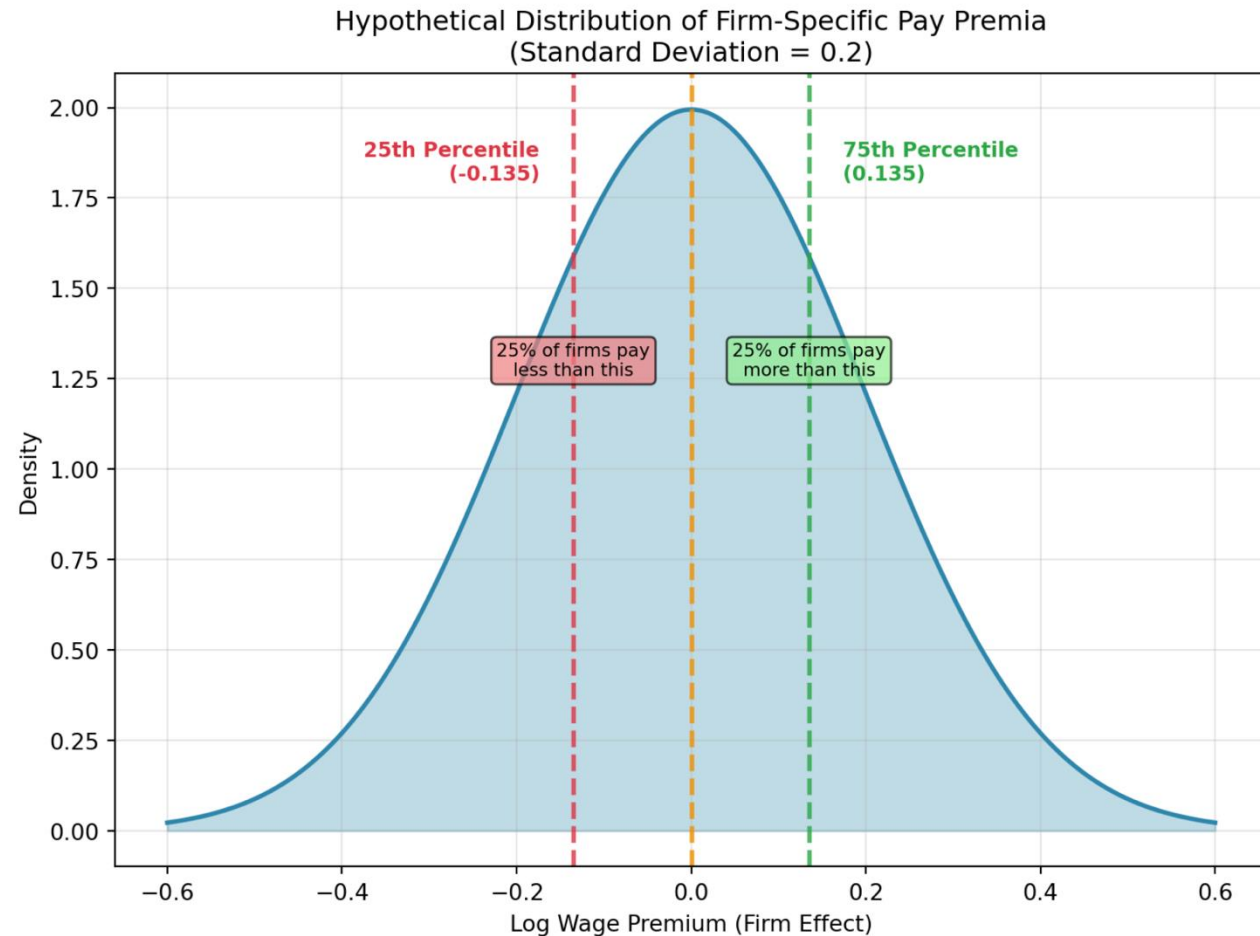
- The fact that some firms pay higher wages than others may seem uncontroversial to you
 - However, once you think about it, it becomes a bit harder to prove
 - Does Google pay more than other firms, or does it attract the highest quality workers who would get paid a lot at other firms as well?
- What we are really interested in whether firms pay different amounts **conditional on the productivity of the worker**

Abowd, Kramarz and Margolis (1999)

- Abowd, Kramarz, and Margolis came up with a solution to this problem, which was later refined by Card, Heining and Kline (2013), among others
- Compute changes in earnings within a worker as they move across firms
- Firms in which workers tend to see pay increases are high-wage firms while firms that tend to see pay decreases are low-wage firm
 - Method allows you to construct a firm-specific pay premia that captures how much a firm pays relative to the average firm

Firm-Specific Pay Premia

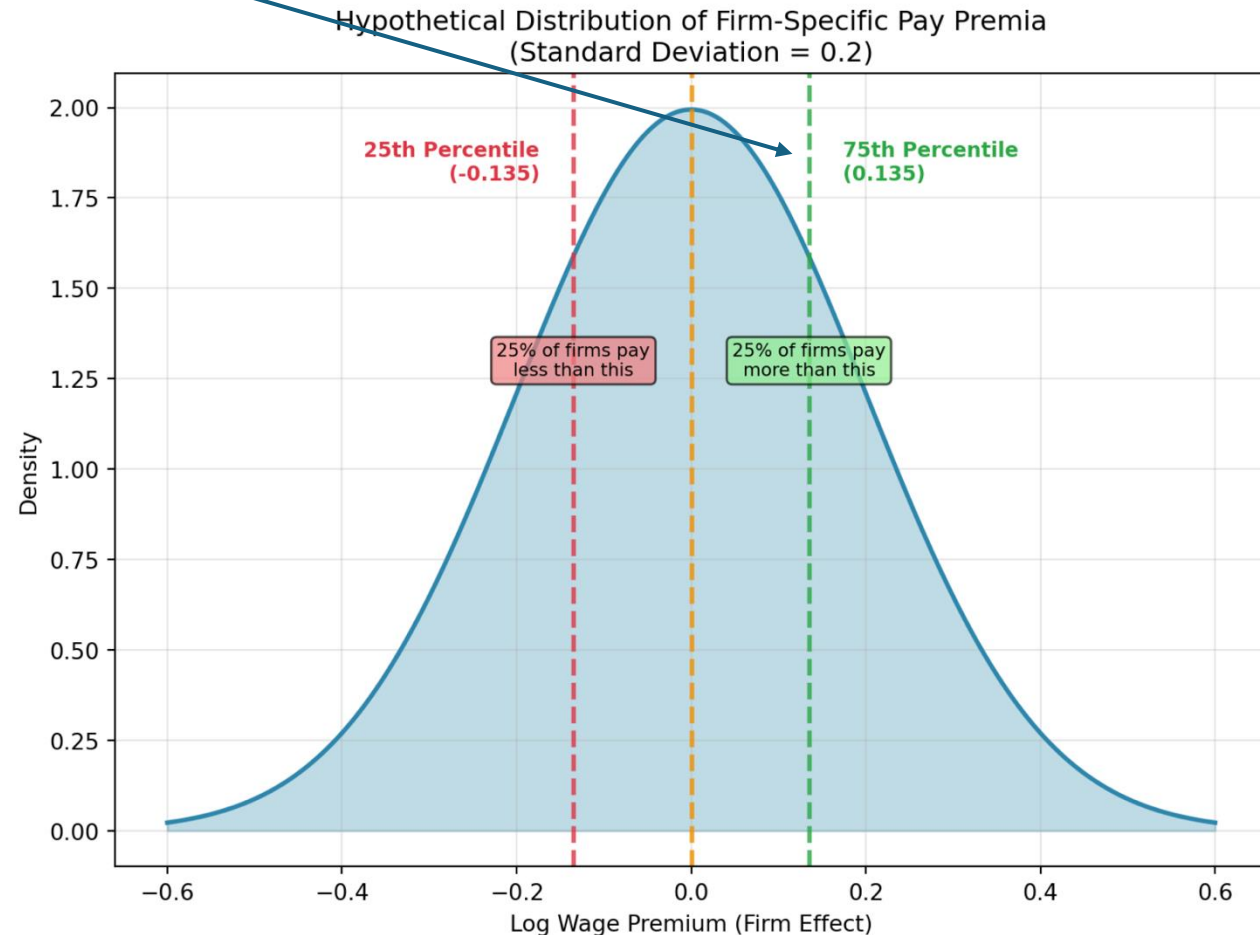
- Card, Heining and Kline (2013) find standard deviation of firm-specific pay premia is equal to 0.2
- What does this actually mean?



Firm-Specific Pay Premia

Moving from average firm to firm at 75th percentile increases wages by about 14.3%

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Job Loss

- In a perfectly competitive model, there is an odd prediction of the impact of job loss on workers

Job Loss

- In a perfectly competitive model, there is an odd prediction of the impact of job loss on workers
- This has no impact on workers, as the perfectly competitive model assumes that workers can without cost transition across firm
- But reality is very different...

Jacobson, LaLonde, and Sullivan (1993)

- Jacobson, LaLonde and Sullivan had an idea to look at how outcomes for workers change after being laid off
 - If workers can find another job quickly, maybe impacts are temporary
- However, there is a problem: what if workers who get laid off are “worse” workers than those who are not laid off
 - This is exactly the same issue as we have discussed previously. There may be a confounding variable that is correlated with both a person’s future earnings potential and whether they are laid off
- To avoid this issue, JLS (1993) focus on mass layoff events (i.e. a firm closing down a branch or division) as a source of firings that are not driven by the quality of the worker

Jacobson, LaLonde, and Sullivan (1993)

Very similar earnings trajectories prior to layoff

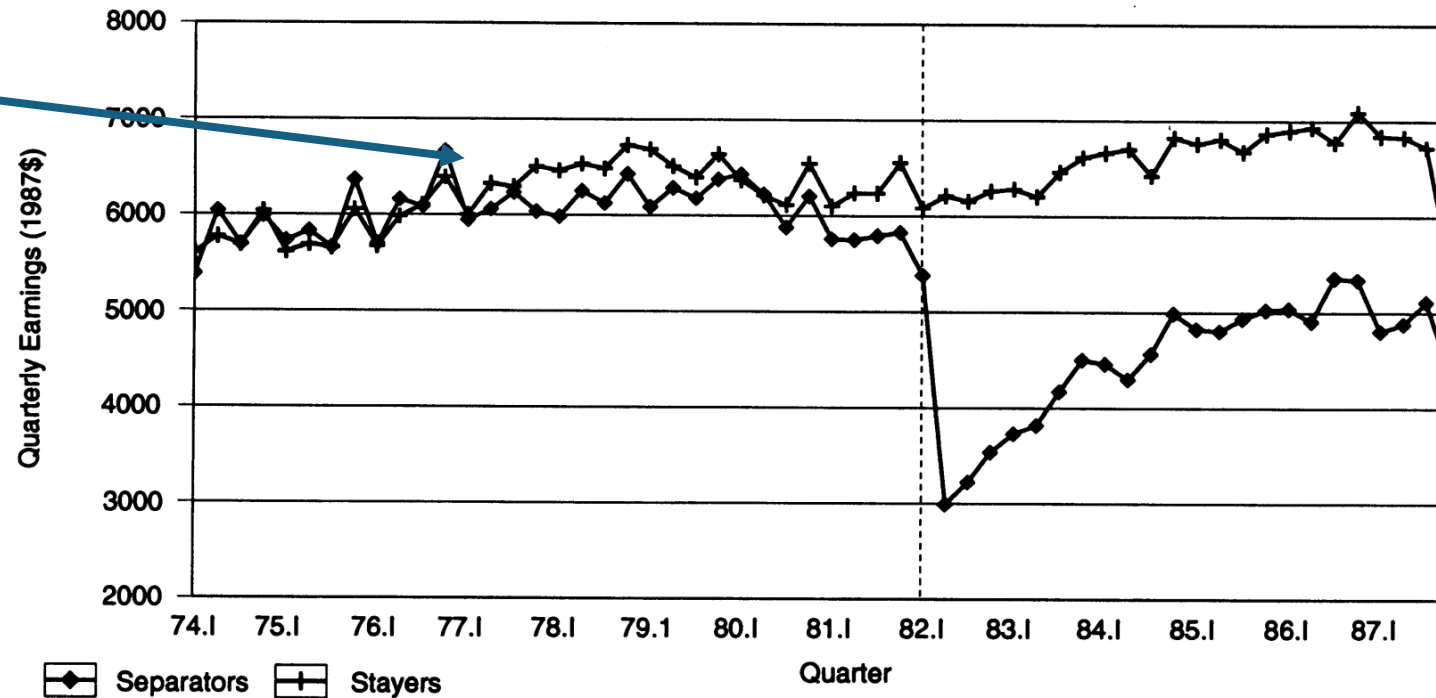


FIGURE 1. QUARTERLY EARNINGS (1987 DOLLARS) OF HIGH-ATTACHMENT WORKERS SEPARATING IN QUARTER 1982:1 AND WORKERS STAYING THROUGH QUARTER 1986:4

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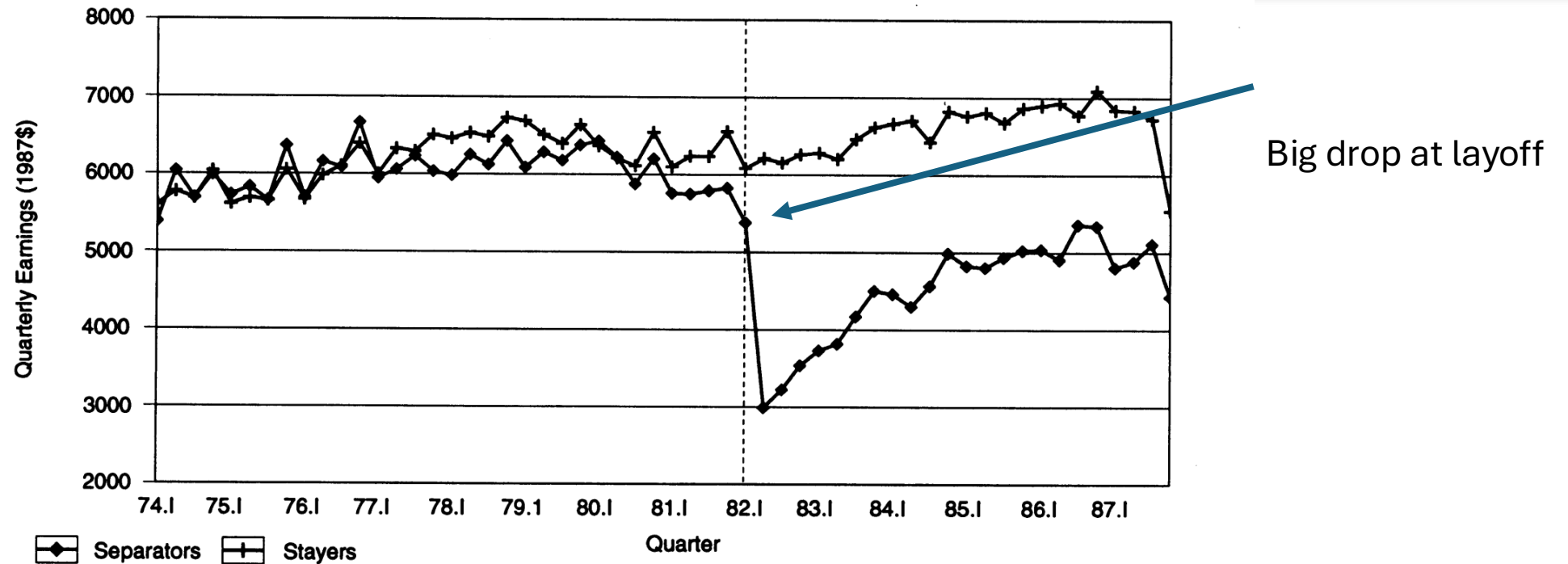


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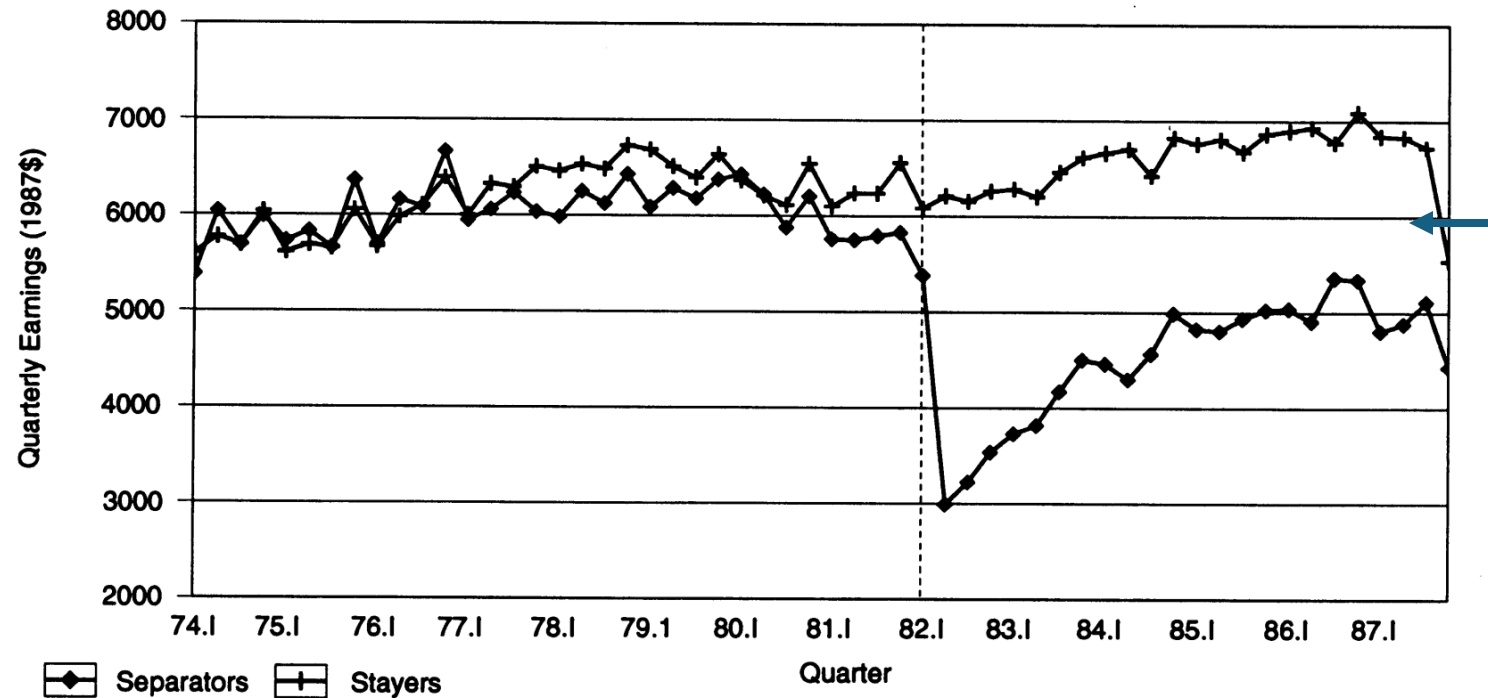
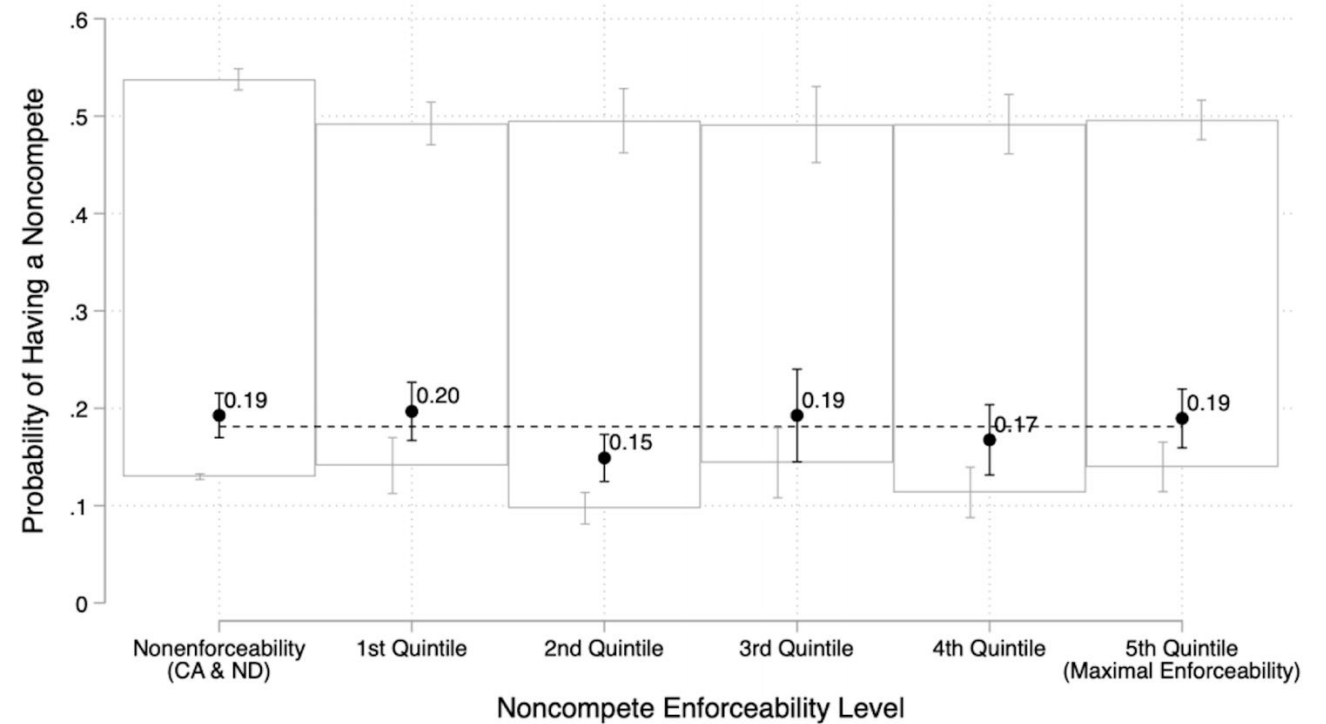


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Institutional Boundaries to Mobility

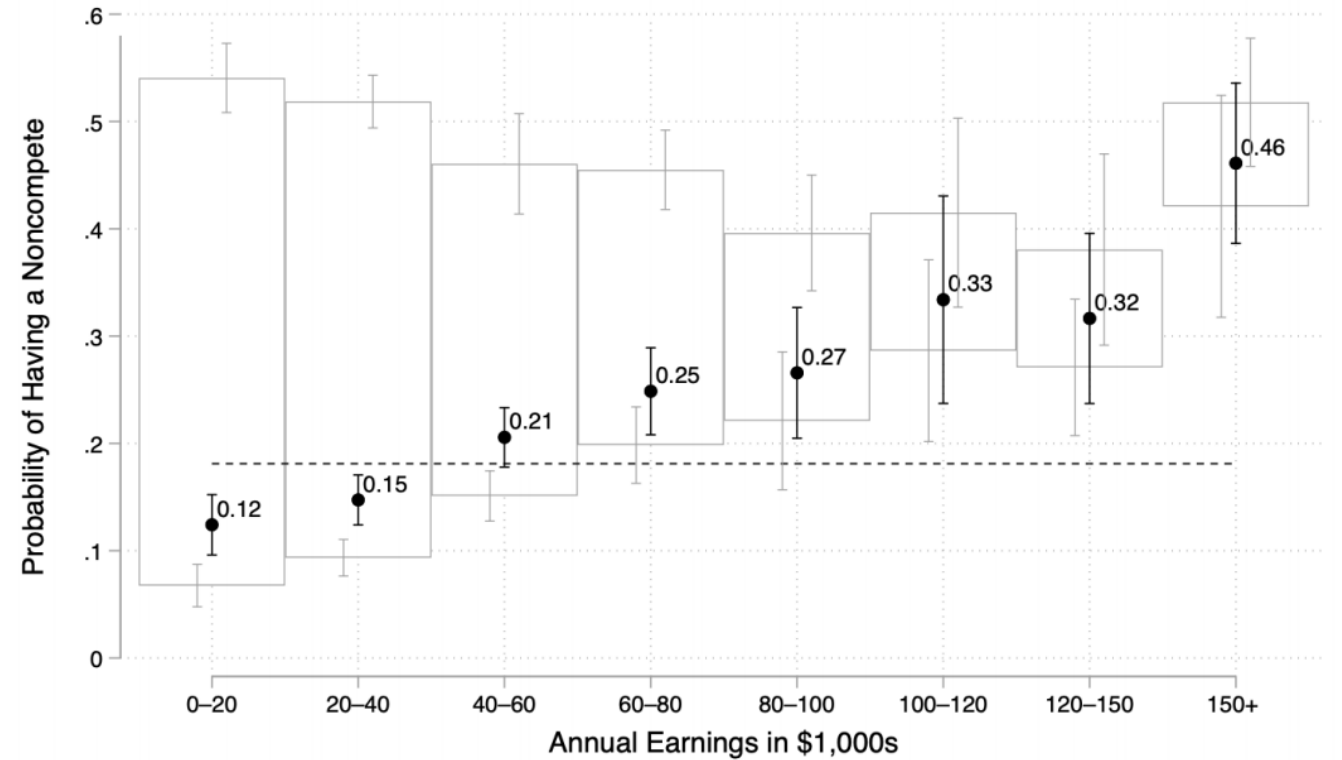
- Noncompete contracts are prevalent in the labor market
 - Noncompetes prevent workers from becoming employed at competitors after leaving their firm
 - Some good theoretical arguments for them: employers may be reluctant to train workers if they can leave for competitors with trade secrets
- But the fact they restrict labor mobility implies they may enhance monopsony power

Institutional Boundaries to Mobility



Noncompetes common even in states in which they are not enforceable.
Source: Starr, Prescott and Bishara (2020)

Institutional Boundaries to Mobility



Noncompetes common even for low-wage workers where they are unlikely to have a legitimate business interest

Institutional Boundaries to Mobility

- FTC attempted to make noncompetes unenforceable nationwide in
- In 2024, this action was blocked by the courts, alleging the FTC was exceeding its authority

No-Poach Agreements at Franchises

- Franchises are generally owned by individuals
- When you become a franchisee, many firms required to sign a no-poach agreement, agreeing not to hire workers from any other franchisee
- Ashenfelter and Krueger (2018) document the extensive use of no-poach agreements

Table 1a: Number of Franchise Chains by Industry and No-Hire Agreement

Industry	No-Poach Agreement		Total
	No	Yes	
Automotive	2	4	6
Baked Goods	1	3	4
Beauty-Related	4	4	8
Business-Related	0	5	5
Child-Related	2	1	3
Decorating & Home Decorating	0	2	2
Education-Related	1	0	1
Frozen Desserts	3	3	6
Health & Fitness	5	6	11
Lodging	12	2	14
Maintenance Services	5	8	13
Personnel Services	0	2	2
Printing	2	0	2
Publications	1	0	1
Quick Service Restaurants	8	32	40
Real Estate	6	1	7
Restaurants (Sit-Do..	0	5	5
Retail Food	3	3	6
Retail Stores	7	4	11
Services-General	2	4	6
Travel	1	2	3
Total	65	91	156

Pearson $\chi^2(20) = 44.6$

P-value = 0.001

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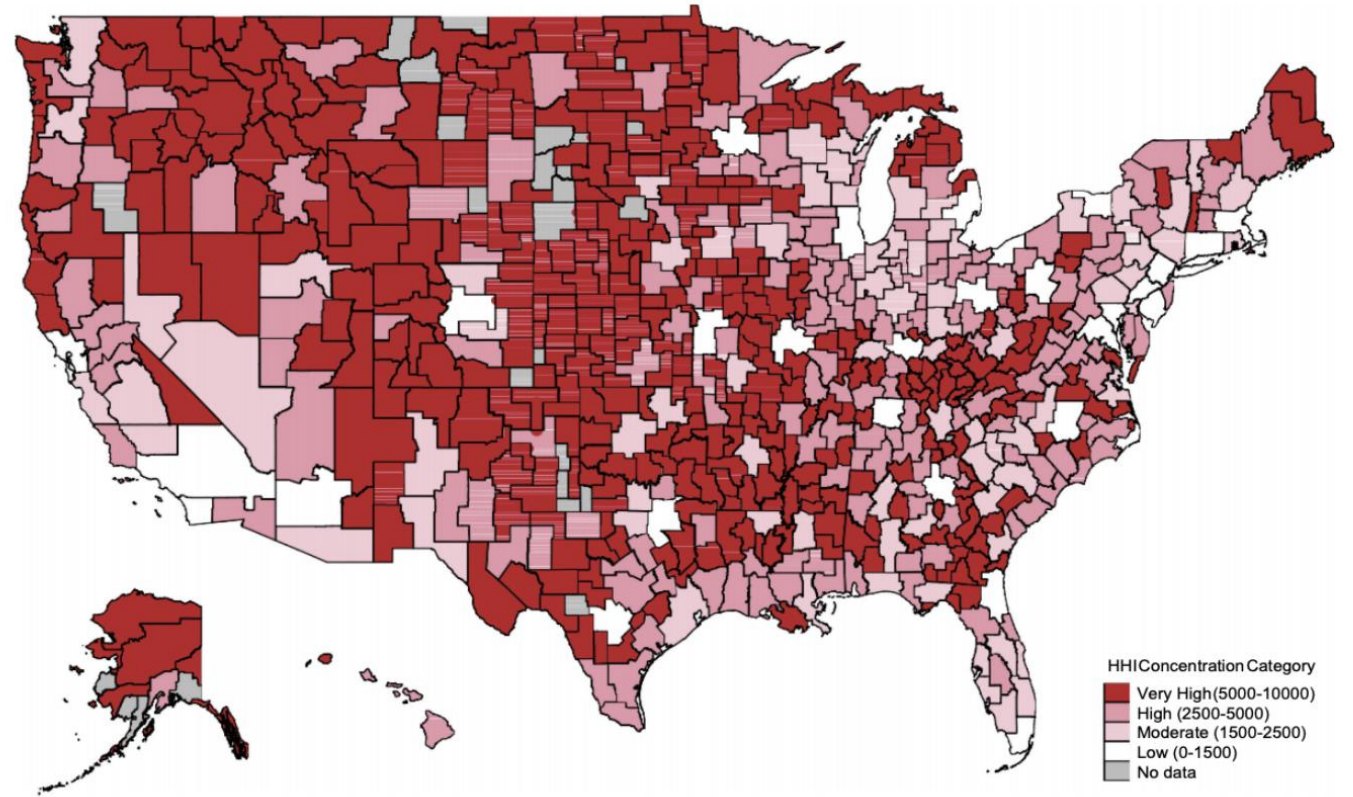
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No-Poach Agreements at Franchises

- Washington State AG launched an investigation of the legality of no-poach agreements
 - This was directly inspired by the Ashenfelter and Krueger (2018) paper
- Franchisees agreed to abandon the use of no-poach agreements going forward

Concentration

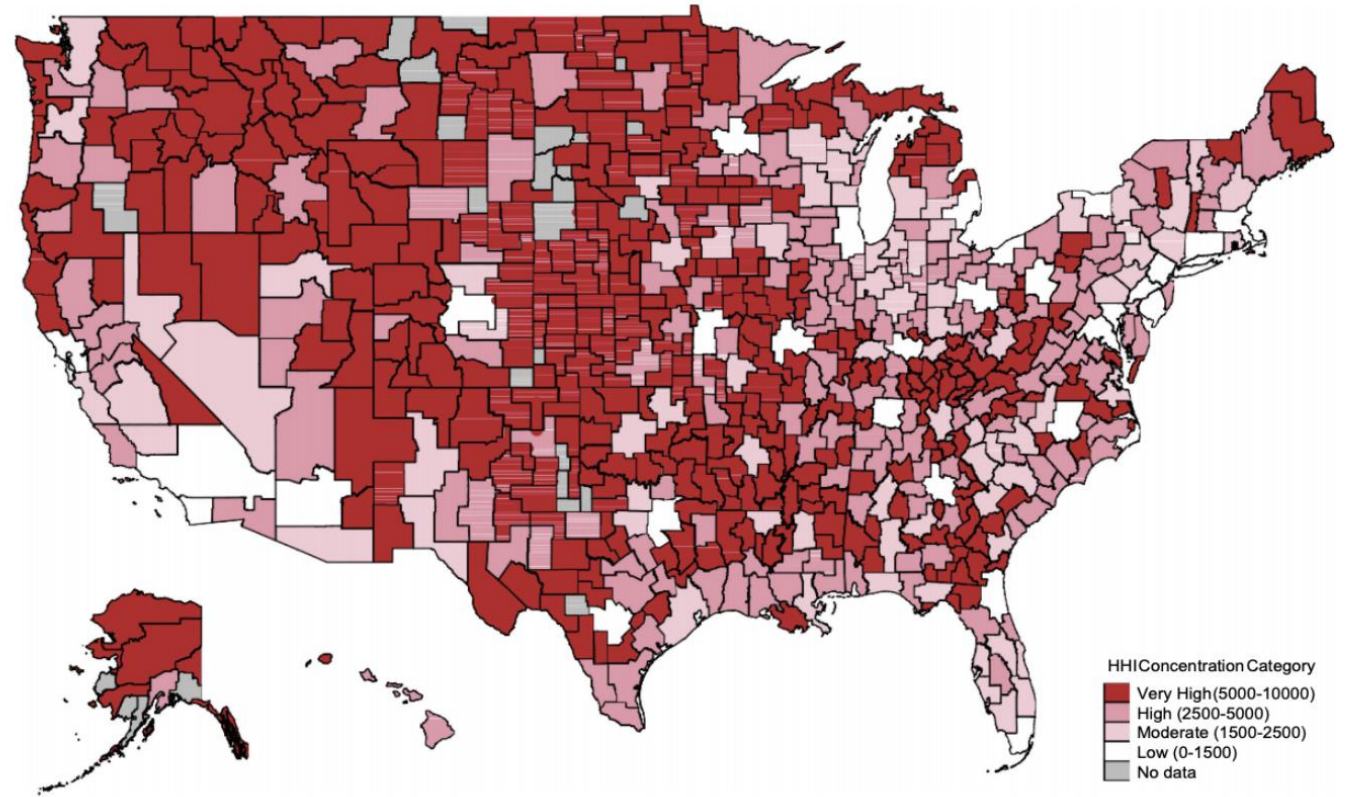
- $HHI = \text{sum of squared market shares}$
- If you define a labor market as an occupation within a commuting zone, you find labor markets tend to be quite concentrated
- A commuting zone is a cluster of counties where counties are grouped together based on commuting flows



Azar, Marinescu and Steinbaum (2022)

Concentration

- Azar, Marinescu and Steinbaum find places with high concentration tend to pay lower wages, consistent with monopsony power
- A common critique, however, again stems from confounding variables
- What if there is some other difference between markets with high concentration and low concentration

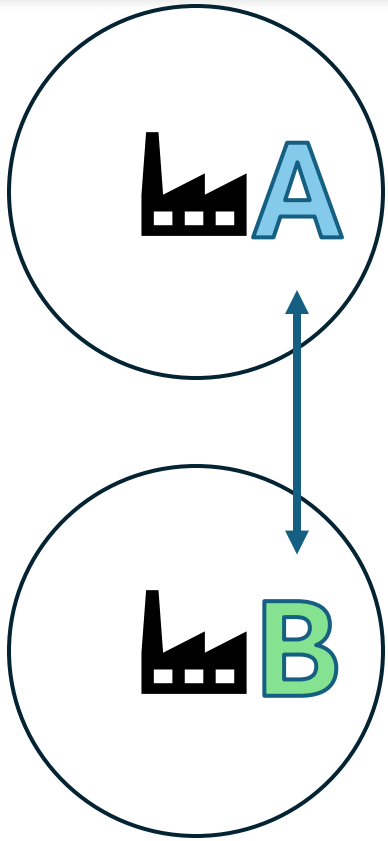


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Mergers

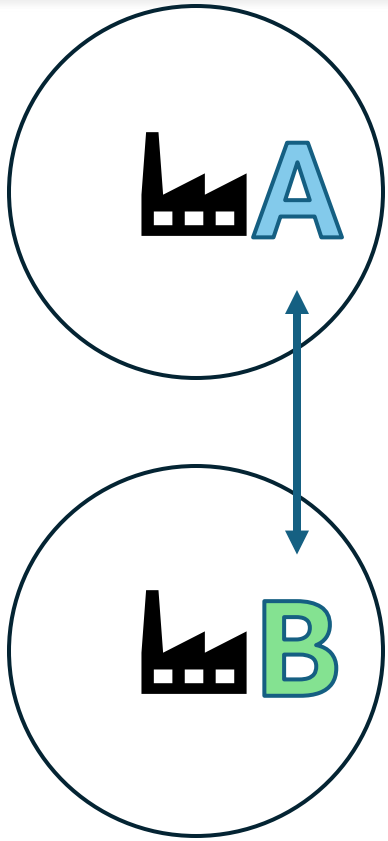
- Mergers offer an opportunity to study concentration changes that are particularly relevant from a policy perspective
 - DOJ/FTC block mergers that potentially decrease competition
 - Also helpful for identification (allows you to study pretrends to confirm something other than concentration is not causing the change)
- But until recently there has been little evidence on whether labor-market power should be a concern for antitrust authorities

Arnold (2025)

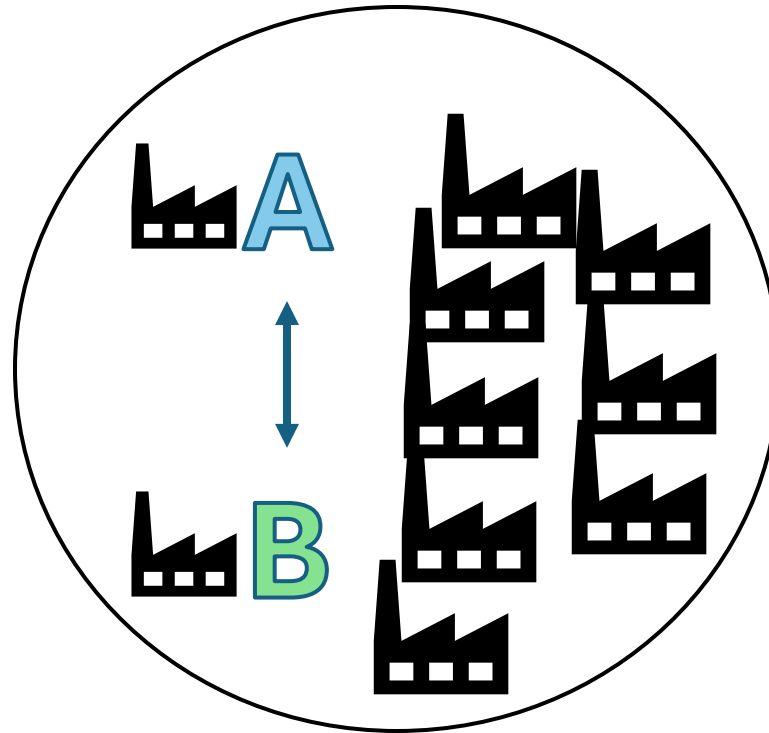


Low-Impact – Firms merging
operate in different
commuting zones

Arnold (2025)



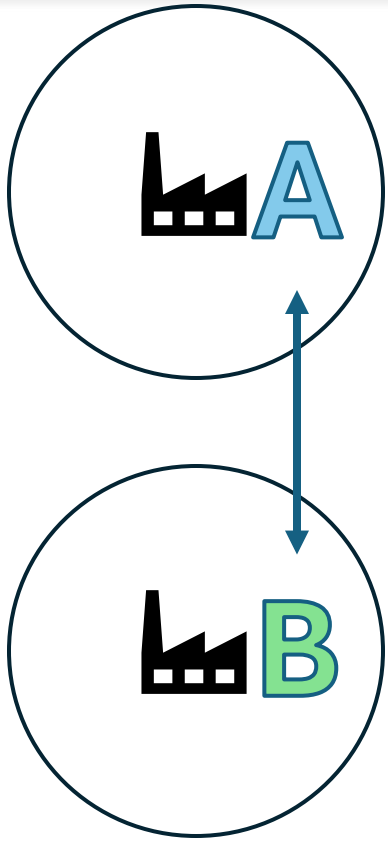
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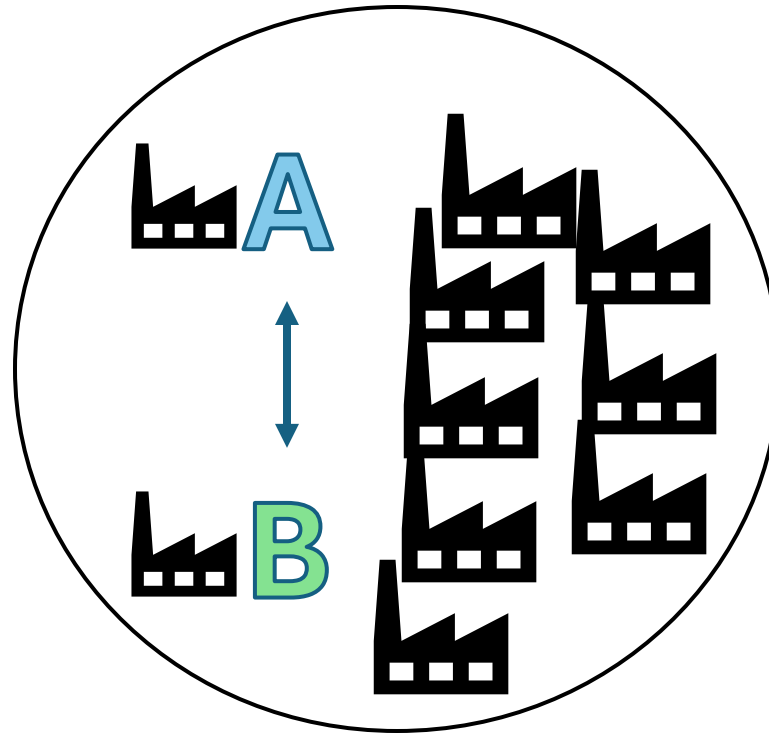
Medium Impact –
Firms merging
operate in same
commuting zone,
but lots of
competitors

Arnold (2025)

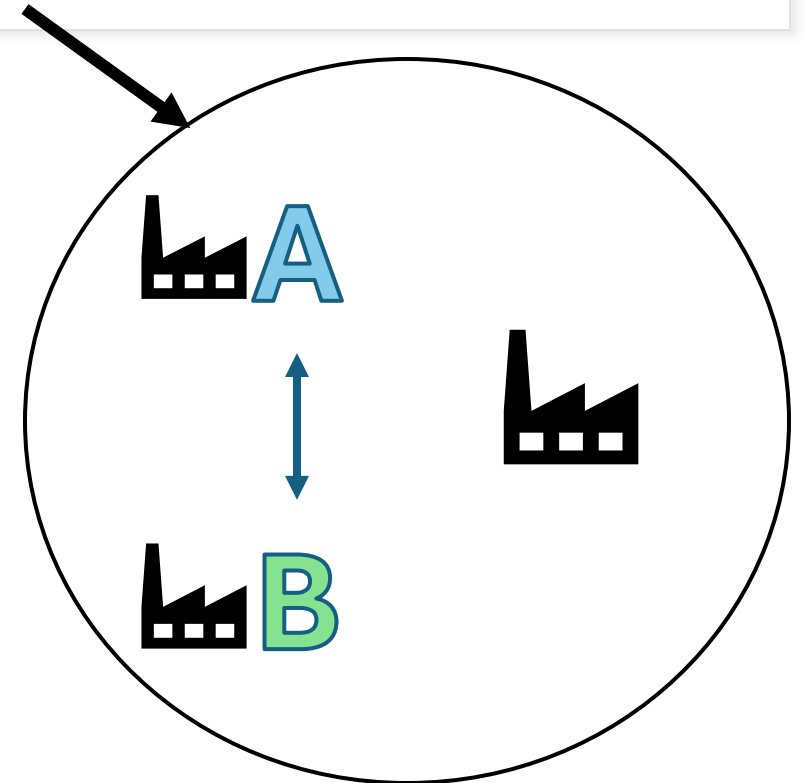
High Impact – Firms merging
operate in same commuting zone
with few competitors



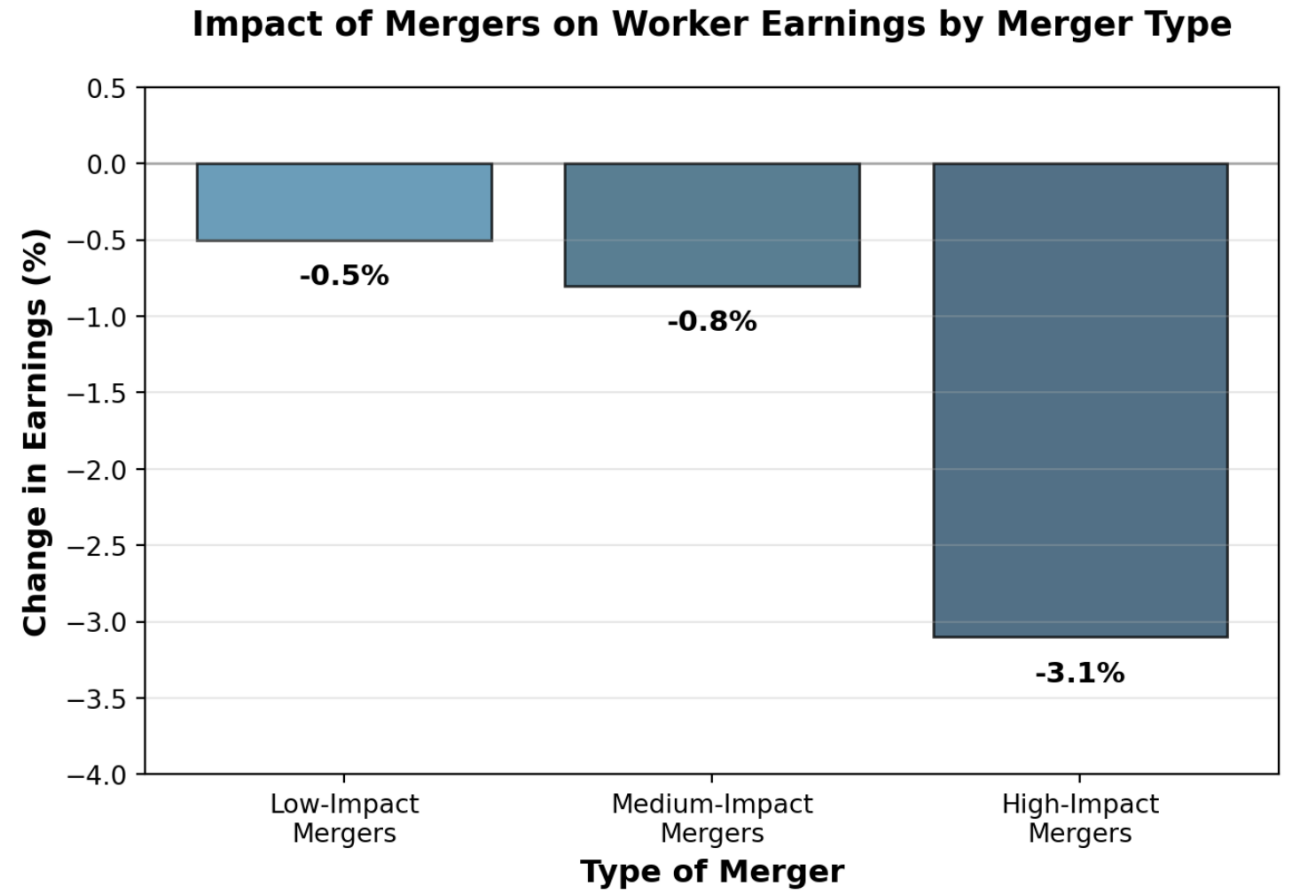
Low-Impact



Medium Impact



Main Result in Arnold (2025)



Albertsons and Kroger Merger

- Until recently labor-market impacts of mergers did not play a role in antitrust scrutiny
- In 2022, Kroger attempted to acquire Albertsons
- These are very large employers (Kroger is the 7th largest employer and Albertsons the 22nd)



Albertsons and Kroger Merger

- In the US, mergers must be submitted to antitrust authorities in order to approved
- Federal Trade Commission (FTC) issued a complaint against the merger, citing probably impacts on grocery prices and **wages of unionized labor**
- First example of labor playing a role in a proposed merger case



Decision

- In the end, the FTC won the case against Kroger and Albertsons, which forced the companies to abandon the deal
- Albertsons is now suing Kroger for breaching the merger agreement and not working with antitrust authorities to reach a deal



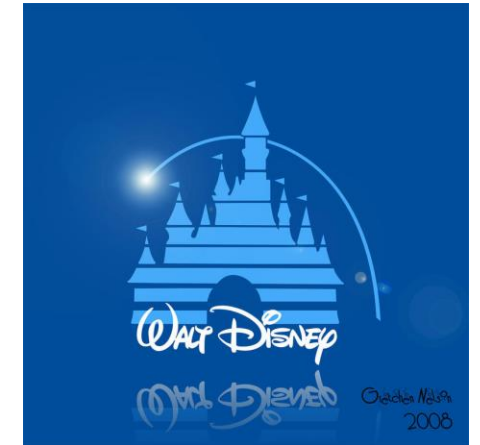
Collusion in Labor Markets

- It is difficult to detect collusion generally, but occasionally cases come to light
- In 2010 a lawsuit was launched against many of the largest employers in Silicon Valley including Apple, Adobe, Google, Intel, Intuit, Lucasfilm, Pixar and eBay
 - The firms had agreed not to cold call each others workers
 - Also included agreements not to counteroffer above initial wage offers, limiting competition



Collusion in Labor Markets

- A similar lawsuit was later launched against many animation studios, including BlueSky, Pixar, Dreamworks, Disney, and LucasFilm
- Again firms were found guilty of agreeing not to poach each others' workers
- Disney now owns many of the companies involved in the lawsuit



Monopsony and Wage Inequality

- In a prior section we studied the evolution of the 90/10 wage ratio
- Sometimes researchers separately study the 90/50 wage ratio and the 50/10 wage ratio
 - 90/50 tells us something about wage inequality at the top (are the high-earners pulling away from the middle)
 - 50/10 tells us something about wage inequality at the bottom (are the lowest-earners falling behind the middle)

Log Wage Ratios

- Sometimes instead of reporting the 90/10 ratio, we report the log of the 90/10 ratio
 - So if 150,000 corresponds to 90th and 50,000 corresponds to 50th, then $\log(150,000/50,000)=\log(3)=1.098$
 - We always use the natural logarithm (i.e. log base e)

Why Logs

- There are a lot of reasons why logs are so common in economics (and math and science generally)
- One reason we are going to use logs is that it gives us a convenient way to decompose earnings inequality into various components
- We are going to use the following property of logarithms

$$\log(A/B) = \log(A) - \log(B)$$

Decomposing Inequality

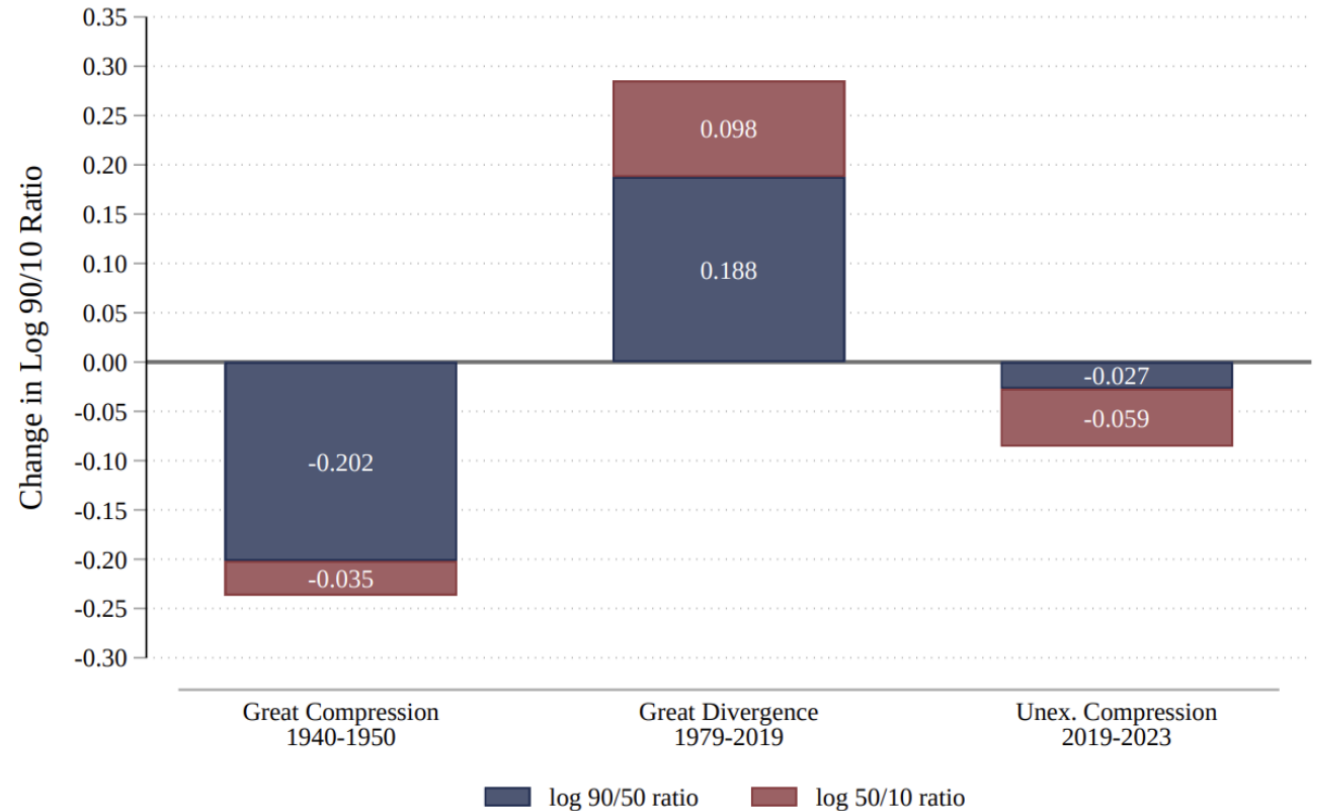
$$\log \left(\frac{W_{90}}{W_{10}} \right) = \log(W_{90}) - \log(W_{10})$$

- If we add and subtract $\log(W_{50})$ we get

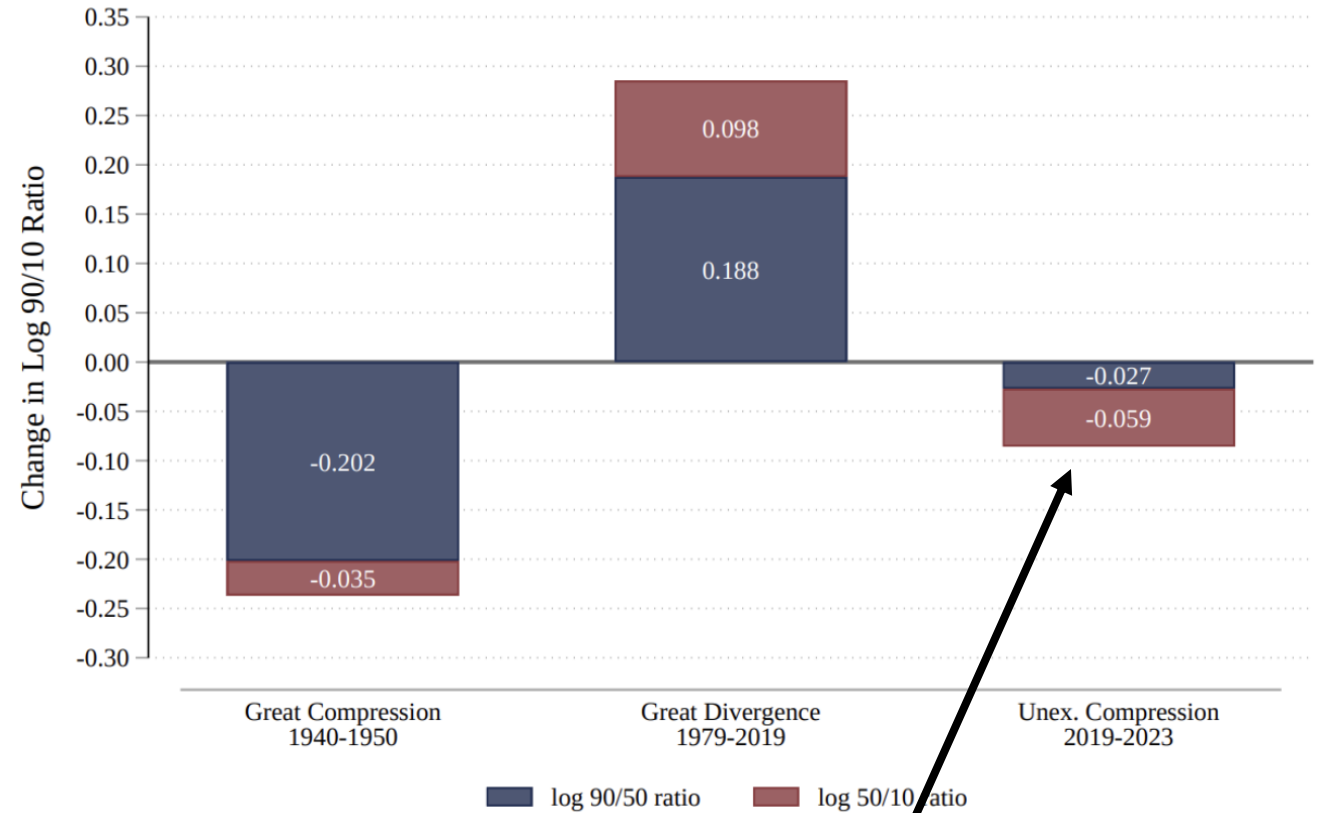
$$\log \left(\frac{W_{90}}{W_{10}} \right) = \underbrace{\log(W_{90}) - \log(W_{50})}_{\text{Upper Tail Inequality}} + \underbrace{\log(W_{50}) - \log(W_{10})}_{\text{Lower Tail Inequality}}$$

Decomposing Inequality

- This comes from Autor, Dube and McGrew (2023)
- We've already discussed these first two episodes
- What is perhaps surprising is that in just 4 years, there has been a rapid decrease in inequality



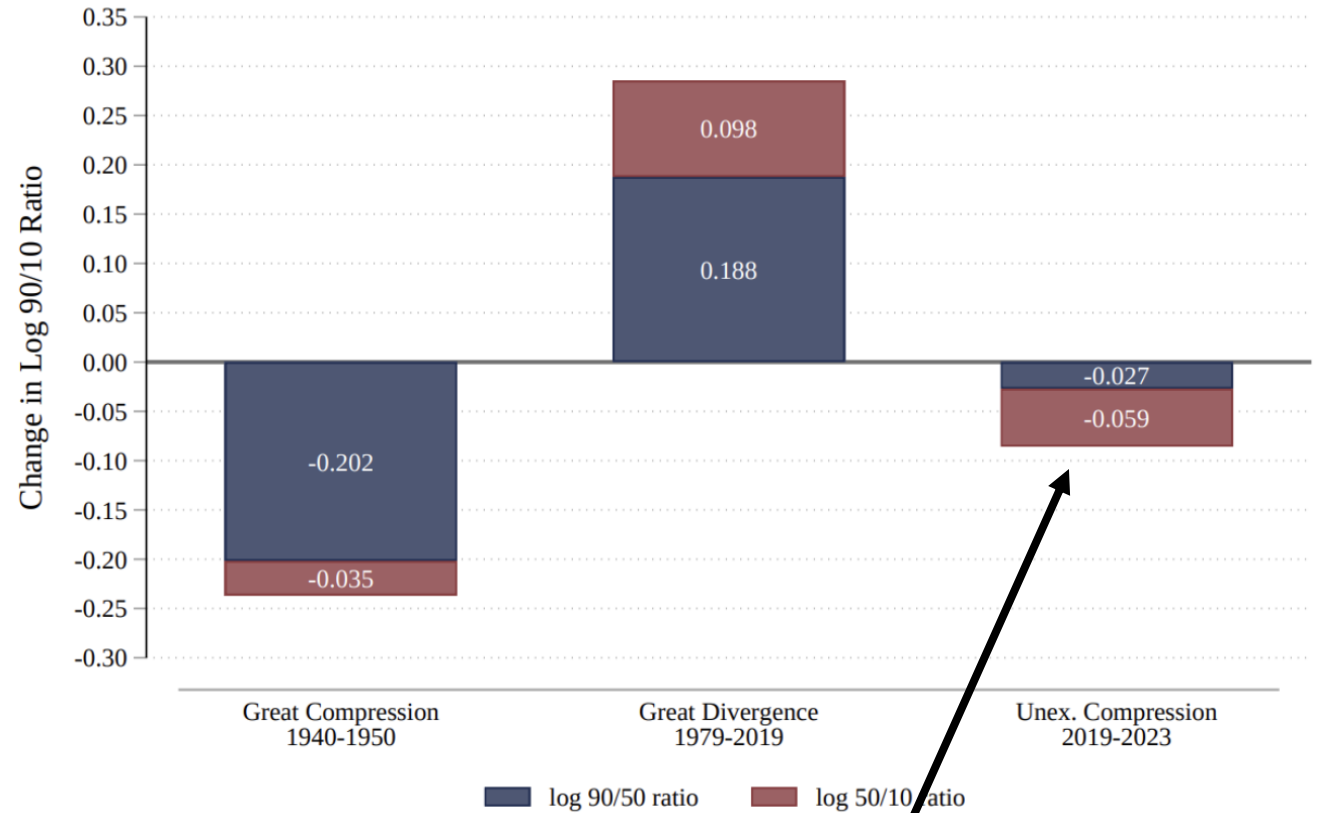
Decomposing Inequality



- Majority of change is coming from the $\log(50/10)$ ratio

Decomposing Inequality

- Could be explained by perfectly competitive model – increase in demand for low-skill work rapidly increased wages
- Autor, Dube, and McGrew (2023) argue that this is unlikely to be the full story. Instead, they argue that monopsony power is key in explaining this compression

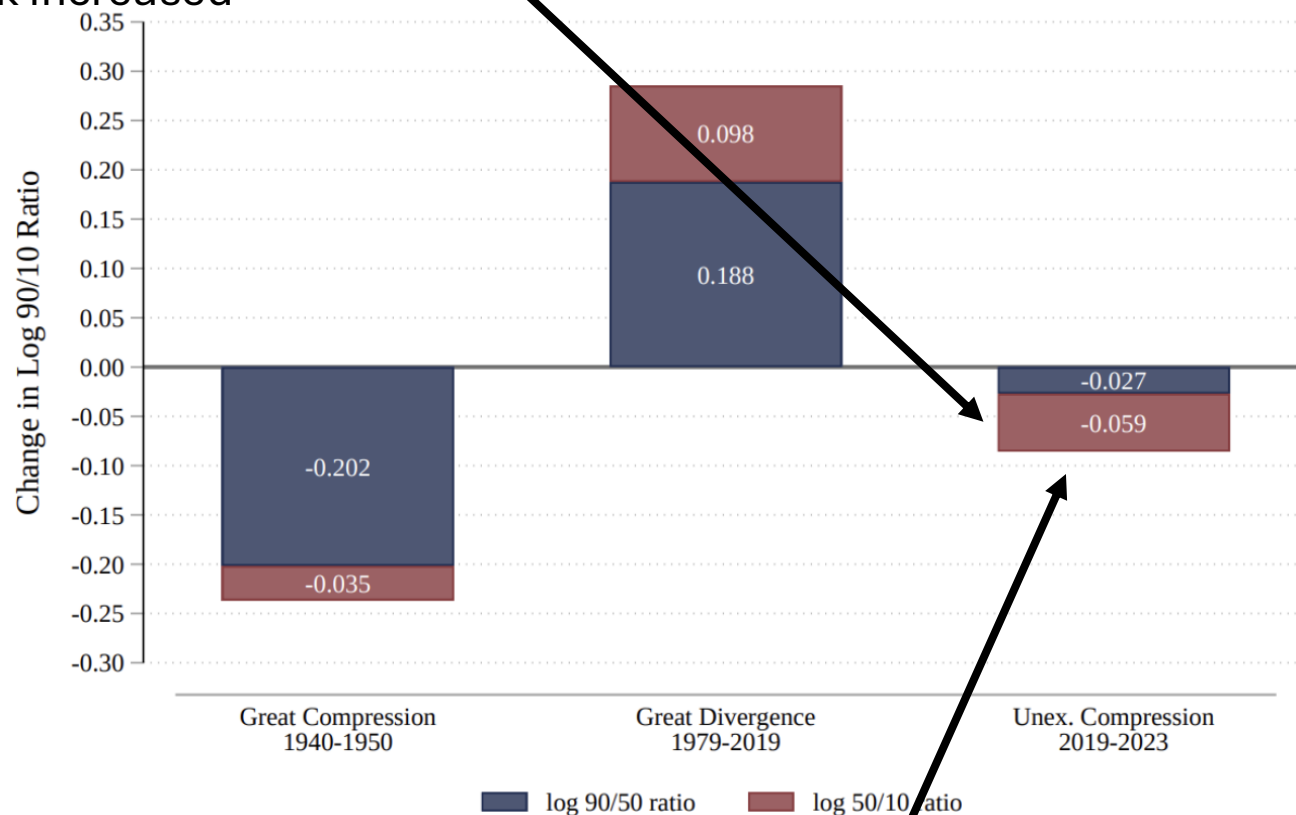


- Majority of change is coming from the $\log(50/10)$ ratio

Changes driven by workers becoming more sensitive to wage differences across firms. This accelerated wage growth as demand for low-skill work increased

Decomposing Inequality

- Could be explained by perfectly competitive model – increase in demand for low-skill work rapidly increased wages
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- Majority of change is coming from the log(50/10) ratio

Monopsony

- Despite the intuitive appeal, monopsony did not catch on for some time
 - For a long time, there was resistance to the idea that monopsony is an important feature of labor markets
- But monopsony models can offer a richer, more realistic picture of the labor market
- When thinking about potential impacts of different policies, important to consider whether a monopsonistic or perfectly competitive model of the labor market is appropriate